

Bayesian inference for active microrheology near a wall

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Abstract

Inference of fluid parameters from the motion of suspended probes is a common approach in microrheology. Near solid boundaries, however, wall-induced hydrodynamic interactions increase the apparent drag on the probe, systematically biasing inferred material properties when left unaccounted for. We develop a data-analysis framework that combines analytical forward models with Bayesian inference to identify parameters of both Newtonian and viscoelastic fluids from the displacement data of a force-controlled sphere translating near a rigid planar wall. For Newtonian fluids, a wall-corrected Stokes drag model is employed to jointly infer the fluid viscosity η_s and the probe-to-wall separation δ from steady-state displacement data. For viscoelastic fluids, this model is extended to a linearized creep-recovery Oldroyd-B formulation retaining the same near-wall corrections, enabling simultaneous estimation of the solvent viscosity η_s , polymer viscosity η_p , relaxation time λ , and wall separation δ , together with uncertainties associated with model bias and measurement error. Using high-fidelity finite-element data as a surrogate for experiments, we demonstrate that the proposed framework accurately recovers the Newtonian viscosity and the three viscoelastic constitutive parameters, together with the wall separation, within the linear viscoelastic regime. Performance deteriorates at large Weissenberg numbers, where posterior predictive checks expose the structural inadequacy of the linear surrogate, turning the inference into a diagnostic for model adequacy rather than only a source of parameter estimates. Beyond the wall, we show that finite periodic-domain interactions bias the inferred modulus unless removed by a Hasimoto correction, after which single-particle probing resolves the local modulus distribution of a heterogeneous viscoelastic material that bulk rheology reports only as an average. The sensitivity of inferred posterior distributions to measurement noise is also quantified, providing insight into the robustness of near-wall microrheological measurements.

Keywords: Bayesian inference, active microrheology, viscoelasticity, wall effects, model discrepancy, Markov chain Monte Carlo

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1. Introduction

Complex fluids are structurally inhomogeneous at nanometer-to-micrometer scales owing to suspended particles, droplets, polymers, or surfactants, and as a consequence exhibit nontrivial rheological behavior with broad scientific and technological relevance [1, 2, 3, 4, 5]. Many biological materials share this complexity, including synovial fluid, mucus, saliva, blood plasma, and cytoplasmic mixtures. Because the mechanical properties of these fluids often reflect the physiological state of the host, the measurement of the response to external stimuli can be used for diagnostic purposes. Examples include elevated extracellular-matrix stiffness at cancer metastatic sites and fibrotic remodeling in cardiac tissue following infarction [6, 7, 8], abnormally viscous mucus in cystic fibrosis [9], and reduced viscoelasticity of synovial fluid in arthritis caused by hyaluronic-acid degradation [10, 11]. Quantifying these properties is therefore important for both biological insight and clinical diagnostics.

Conventional bulk rheometers measure spatially averaged properties and cannot resolve local mechanical heterogeneity at cellular or subcellular length scales, which are required for many diagnostic applications [12, 13, 14, 15]. Microrheology addresses this limitation by probing at the microscale. Passive methods infer viscoelasticity from thermal fluctuations [16, 17, 18], whereas active methods drive particles using magnetic fields or optical tweezers [19, 20, 21, 22, 23]. Active microrheology is particularly suited to stiffer or more viscous media, where thermal fluctuations are too small to induce measurable Brownian displacement [24, 25]. Material properties are inferred by fitting measured probe displacements to analytical models based on classical relations such as Stokes drag or generalized Stokes–Einstein relations. These models typically assume motion in an unbounded fluid. In practice, however, proximity to solid boundaries introduces wall-induced hydrodynamic interactions that influence the drag on the probe, systematically biasing the inferred properties when unaccounted for [26]. Although experimental validations are often performed far from walls [27, 28, 23], real biological samples cannot avoid solid boundaries, especially when the sample volume is small or when measurements are performed adjacent to tissue.

A further complication is that microrheological data typically exhibit considerable scatter, arising from variability in applied forces, particle surface properties, optical aberrations, post-processing, and thermal noise. These variations are most commonly handled by simple linear regression. For nonlinear viscoelastic fluids, however, linear regression can itself introduce systematic measurement error. Moreover, explicit tracking of the probe-to-wall distance, while attempted in several studies [29], places stringent demands on optical resolution and data quality, introducing additional uncertainty. A rigorous statistical framework that propagates all sources of uncertainty into the inferred parameters is therefore needed. Bayesian inference

provides such a framework and has already proved useful for identifying viscoelastic material parameters, interpreting optical-tweezer microrheology data, and calibrating rheological models in complex flow geometries [30, 31, 32, 33]. These studies show that uncertainty quantification is most valuable when the forward model, the measurement process, and the limits of model validity are treated as part of one inference problem. Unlike these earlier applications, which do not address a nearby boundary, the present work infers the probe-to-wall separation jointly with the constitutive parameters and uses posterior predictive checks to detect when the linearized surrogate ceases to describe the data, turning the Bayesian framework into a diagnostic for model adequacy rather than only a source of parameter uncertainties.

We develop a data-analysis framework that combines analytical forward models with Bayesian inference to extract rheological information from the displacement of a force-controlled sphere translating at a fixed angle near a planar rigid wall. For Newtonian fluids, a closed-form displacement model based on Stokes drag with direction-dependent wall-correction factors enables joint inference of the fluid viscosity η_s and the probe-to-wall separation δ . For viscoelastic fluids, this model is extended to a linearized, creep-recovery Oldroyd-B formulation retaining the same wall corrections, enabling simultaneous inference of η_s , η_p , λ , and δ , together with fully quantified parameter uncertainties. High-fidelity finite-element solutions serve as surrogate experimental data throughout.

The paper is organized as follows. Section 2 introduces the Bayesian inverse problem and the observation model used to separate measurement noise from model bias. Section 3 states the full-order fluid-particle model used to generate reference trajectories, and Section 4 derives the analytical reduced-order models used inside the inference. Section 5 then specifies the prior distributions, likelihood, and Markov chain Monte Carlo (MCMC) sampler. With these ingredients in place, Section 6 reports numerical experiments for Newtonian and viscoelastic fluids, including a high-Weissenberg-number stress test that exposes model inadequacy through posterior predictive checks.

2. Bayesian inference framework

A central objective in microrheology is to make reliable inferences about the rheological properties of a fluid from the motion of an embedded probe. Near a solid boundary, this objective is complicated by two distinct sources of error. The forward model, which maps a candidate parameter vector to a predicted probe trajectory, is an analytical idealization of the underlying physics and is therefore subject to model bias. The displacement data, in turn, are corrupted by experimental noise originating from particle-tracking uncertainty, optical aberrations, post-processing artifacts, and thermal fluctuations. A deterministic least-

squares fit treats both contributions as a single residual to be minimized and returns a point estimate whose accuracy is difficult to assess. Bayesian inference provides an alternative formulation in which the unknown parameters are modeled as random variables described by probability distributions, and the discrepancy between the model and the data is explicitly decomposed into experimental noise and model bias.

In this work, the Bayesian framework is used exclusively for parameter identification. We develop two separate analytical forward models, one for Newtonian fluids and one for linear viscoelastic fluids, and the inference returns the posterior density of their parameters given the observed trajectory. The forward models themselves are constructed in Section 4; the prior distributions, likelihood, and Markov chain Monte Carlo (MCMC) sampler are specified in Section 5.

2.1. Observation model

We consider a measured trajectory $\mathbf{y}_{\text{obs}} = \{\mathbf{y}_{\text{obs},1}, \dots, \mathbf{y}_{\text{obs},N}\}$ of N probe displacements, where $\mathbf{y}_{\text{obs},i} \in \mathbb{R}^2$ collects the wall-parallel and wall-normal components at time t_i . Let $\mathbf{g}_i$ denote the realization of physical reality corresponding to the i th observation, and let $\mathbf{d}_i(\boldsymbol{\mu}) \in \mathbb{R}^2$ denote the prediction of the forward model at t_i for the parameter vector $\boldsymbol{\mu} \in \boldsymbol{\Theta}$. The forward model is the deterministic mapping $\mathbf{d} : \boldsymbol{\Theta} \rightarrow \mathbb{R}^{2N}$ that returns the predicted displacements at the N measurement times.

The experimental error $\boldsymbol{\varepsilon}_{\text{exp},i} \in \mathbb{R}^2$ between the ground truth and the measurement is defined as

$$\boldsymbol{\varepsilon}_{\text{exp},i} = \mathbf{y}_{\text{obs},i} - \mathbf{g}_i, \quad (1)$$

and the model bias $\boldsymbol{\varepsilon}_{\text{bias},i}(\boldsymbol{\mu}) \in \mathbb{R}^2$ between the ground truth and the model prediction as

$$\boldsymbol{\varepsilon}_{\text{bias},i}(\boldsymbol{\mu}) = \mathbf{g}_i - \mathbf{d}_i(\boldsymbol{\mu}). \quad (2)$$

The ground truth $\mathbf{g}_i$ is unknown in practice. Adding Eqs. (1) and (2) eliminates it from the error definition,

$$\mathbf{y}_{\text{obs},i} = \mathbf{d}_i(\boldsymbol{\mu}) + \boldsymbol{\varepsilon}_{\text{exp},i} + \boldsymbol{\varepsilon}_{\text{bias},i}(\boldsymbol{\mu}), \quad (3)$$

which formalizes that the mismatch between model prediction and observation originates from two sources: measurement noise, captured by $\boldsymbol{\varepsilon}_{\text{exp}}$ and assumed to be independent of $\boldsymbol{\mu}$; and model inadequacy, captured by $\boldsymbol{\varepsilon}_{\text{bias}}(\boldsymbol{\mu})$ and depending on $\boldsymbol{\mu}$ through the forward map. In many studies, these two contributions are combined into a single effective noise model. In this work, following [34], they are kept separate so that

the parameter-dependent systematic discrepancy and the parameter-independent measurement noise can be inferred jointly with $\boldsymbol{\mu}$ and reported as distinct quantities.

2.2. Bayesian inference of parameters

Given the observation model (3), the parameter identification problem is posed in the Bayesian setting. Bayes' theorem expresses the probability density of $\boldsymbol{\mu}$ given the observed trajectory $\mathbf{y}_{\text{obs}}$, referred to as the posterior, as

$$p(\boldsymbol{\mu} \mid \mathbf{y}_{\text{obs}}) = \frac{p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\mu}) p(\boldsymbol{\mu})}{p(\mathbf{y}_{\text{obs}})} = \frac{\mathcal{L}(\boldsymbol{\mu} \mid \mathbf{y}_{\text{obs}}) p(\boldsymbol{\mu})}{p(\mathbf{y}_{\text{obs}})}, \quad (4)$$

where $\mathcal{L}(\boldsymbol{\mu} \mid \mathbf{y}_{\text{obs}}) \equiv p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\mu})$ is the likelihood, $p(\boldsymbol{\mu})$ the prior, and

$$p(\mathbf{y}_{\text{obs}}) = \int_{\Theta} \mathcal{L}(\boldsymbol{\mu} \mid \mathbf{y}_{\text{obs}}) p(\boldsymbol{\mu}) d\boldsymbol{\mu} \quad (5)$$

the evidence.

The prior $p(\boldsymbol{\mu})$ encodes information available before the trajectory is observed, such as positivity constraints, expert knowledge, or physical intuition regarding the individual parameters or independent measurements. The likelihood $\mathcal{L}(\boldsymbol{\mu} \mid \mathbf{y}_{\text{obs}})$ quantifies how plausibly a parameter $\boldsymbol{\mu}$ explains the observed trajectory; through the observation model (3), both the measurement noise and the model bias enter the inference via this term.

The evidence $p(\mathbf{y}_{\text{obs}})$ normalizes the posterior over Θ , but its evaluation requires an integral over the entire parameter domain, which becomes computationally prohibitive in high dimensions. The posterior is therefore characterized using a Markov chain Monte Carlo (MCMC) sampler, which generates samples from $p(\boldsymbol{\mu} \mid \mathbf{y}_{\text{obs}})$ using only ratios of posterior densities, in which the evidence cancels.

The posterior combines the prior with the information contributed by the data through the likelihood. A sharp, informative prior dominates the posterior when the data are sparse or noisy. In contrast, a broad prior is overwhelmed by the likelihood when the dataset is large or strongly informative. From the posterior, one extracts posterior means as point estimates, credible intervals describing the spread of plausible parameter values, and the correlation structure between inferred parameters. For the nonlinear forward models considered here, the posterior admits no closed form and is characterized numerically, as detailed in Section 5.

3. Full-order model

This section defines the full-order model used to generate the high-fidelity particle trajectories that serve as reference data throughout the paper. We first specify the geometry and force-controlled particle motion, then state the incompressible creeping-flow equations together with the Newtonian and viscoelastic constitutive laws. The fluid-particle problem is then closed by the boundary and force-balance conditions, followed by a summary of the finite-element discretization, ALE mesh motion, and stabilization used in the simulations.

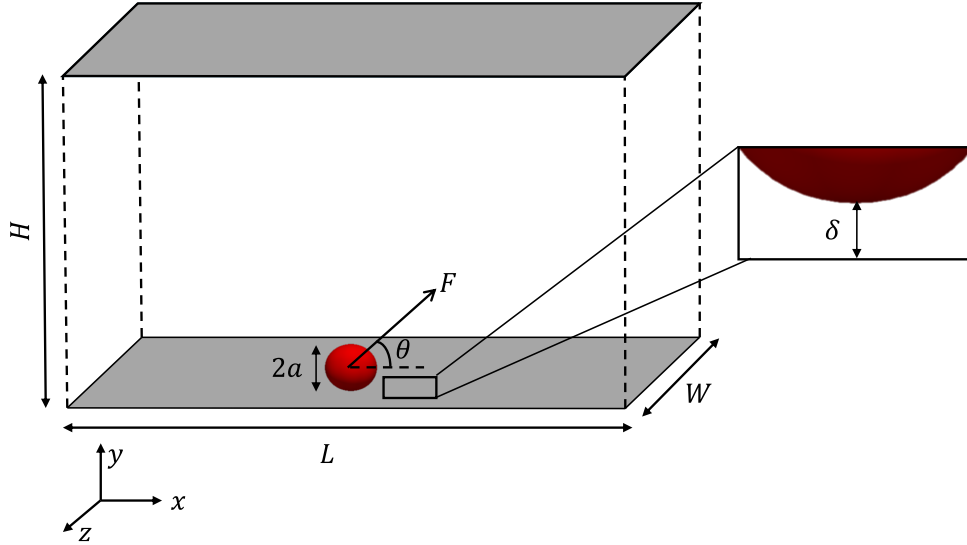


Figure 1: Schematic of a rigid sphere of radius a suspended in a complex fluid confined by a planar rigid wall. The shaded region denotes the wall boundary. The external force $\mathbf{F}$ is applied at an angle θ measured from the wall-parallel direction and δ denotes the surface-to-wall gap.

3.1. Problem definition

We consider a rigid spherical particle P of radius a translating in a fluid domain $\Omega \setminus P$ near a planar rigid wall. The particle is initially separated from the wall by a surface-to-wall gap δ . The flow is incompressible and inertialess, and motion is driven by a constant external force $\mathbf{F}$ of magnitude $F = \|\mathbf{F}\|$ applied to the particle at a fixed angle θ with respect to the wall-parallel direction,

$$\mathbf{F} = (F \cos \theta, F \sin \theta, 0), \quad (6)$$

so that $\theta = 0^\circ$ corresponds to purely wall-parallel forcing and $\theta = 90^\circ$ to purely wall-normal forcing. The geometry, coordinate system, and forcing angle are shown in Fig. 1. The domain dimensions L , W ,

and H , together with the mesh-convergence study supporting the spatial discretization, are reported in Section Appendix A.

3.2. Governing equations

In the creeping-flow regime, fluid motion is governed by the momentum balance and the incompressibility constraint,

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0} \quad \text{in } \Omega \setminus P, \quad (7)$$

$$\nabla \cdot \mathbf{u} = 0 \quad \text{in } \Omega \setminus P, \quad (8)$$

where $\boldsymbol{\sigma}$ denotes the Cauchy stress tensor and $\mathbf{u}$ the fluid velocity field. To close Eq. (7), we use either a Newtonian constitutive law or a viscoelastic Giesekus model.

For a Newtonian fluid,

$$\boldsymbol{\sigma} = -p \mathbf{I} + 2\eta_s \mathbf{D}, \quad (9)$$

where p is the pressure, $\mathbf{I}$ is the identity tensor, η_s is the solvent viscosity, and $\mathbf{D} = \frac{1}{2}[\nabla \mathbf{u} + (\nabla \mathbf{u})^T]$ is the rate-of-deformation tensor. Throughout, the velocity gradient is taken with the index convention $(\nabla \mathbf{u})_{ij} = \partial u_i / \partial x_j$, which fixes the placement of the transpose in the upper-convected derivative below.

For the viscoelastic cases, the total Cauchy stress is decomposed into a Newtonian solvent contribution and a polymeric extra-stress tensor $\boldsymbol{\tau}_p$,

$$\boldsymbol{\sigma} = -p \mathbf{I} + 2\eta_s \mathbf{D} + \boldsymbol{\tau}_p. \quad (10)$$

The polymeric stress evolves according to the Giesekus constitutive equation [35, 36],

$$\lambda \overset{\nabla}{\boldsymbol{\tau}}_p + \boldsymbol{\tau}_p + \frac{\lambda \alpha}{\eta_p} \boldsymbol{\tau}_p \cdot \boldsymbol{\tau}_p = 2\eta_p \mathbf{D} \quad \text{in } \Omega \setminus P, \quad (11)$$

where $\lambda > 0$ is the relaxation time, $\eta_p > 0$ the polymer viscosity, and $\alpha \in [0, 0.5]$ the mobility parameter that controls shear-thinning. The Oldroyd-B model is recovered in the limit $\alpha = 0$ [37].

The upper-convected derivative appearing in Eq. (11) is defined by

$$\overset{\nabla}{\boldsymbol{\tau}}_p = \frac{D\boldsymbol{\tau}_p}{Dt} - (\nabla \mathbf{u})^T \cdot \boldsymbol{\tau}_p - \boldsymbol{\tau}_p \cdot \nabla \mathbf{u}, \quad (12)$$

where D/Dt denotes the material derivative.

For the numerical implementation, the same constitutive law is written in terms of the symmetric positive-definite conformation tensor $\mathbf{c}$, related to $\boldsymbol{\tau}_p$ by

$$\boldsymbol{\tau}_p = G(\mathbf{c} - \mathbf{I}), \quad G = \frac{\eta_p}{\lambda}. \quad (13)$$

Substitution into Eq. (11) gives the conformation-tensor form used in the finite-element calculations,

$$\lambda \overset{\nabla}{\mathbf{c}} + (\mathbf{c} - \mathbf{I}) + \alpha (\mathbf{c} - \mathbf{I})^2 = \mathbf{0} \quad \text{in } \Omega \setminus P. \quad (14)$$

3.3. Boundary conditions

The field equations (7)–(14) are closed by velocity conditions on the particle and on the exterior boundaries. For wall-bounded simulations, the fixed exterior boundaries are no-slip walls, so that

$$\mathbf{u} = \mathbf{0} \quad \text{on } \partial\Omega. \quad (15)$$

The no-slip condition on the particle surface enforces the rigid-body motion of the sphere,

$$\mathbf{u} = \mathbf{U} + \boldsymbol{\omega}_p \times \mathbf{r} \quad \text{on } \partial P, \quad (16)$$

where $\mathbf{U}$ is the translational velocity of the particle center, $\boldsymbol{\omega}_p$ its angular velocity, and $\mathbf{r}$ the position vector measured from the particle center. The motion is force-controlled: the applied force $\mathbf{F}$ is prescribed by Eq. (6), whereas $\mathbf{U}$ and $\boldsymbol{\omega}_p$ are solved as part of the coupled fluid-particle problem.

In the absence of inertia and body forces, the prescribed external force $\mathbf{F}$ acting on the particle is balanced by the surface integral of the hydrodynamic traction,

$$\mathbf{F} = \int_{\partial P} \boldsymbol{\sigma} \cdot \mathbf{n} \, dS, \quad (17)$$

where $\mathbf{n}$ is the outward unit normal on ∂P pointing into the fluid. With this normal convention, Eq. (17) provides the closure that determines the unknown particle velocities $\mathbf{U}$ and $\boldsymbol{\omega}_p$ from the prescribed force.

Two geometries are considered. For a particle near a single wall, we adopt Cartesian axes in which x is wall-parallel, y is wall-normal, and z is the remaining out-of-plane direction; the wall coincides with a plane of constant y , and the applied force (6) lies in the x - y plane. Because neither the geometry nor the forcing

has any z -component, the problem is symmetric about the out-of-plane mid-plane $z = 0$ through the particle center. This is a symmetry of the flow, not the wall: it leaves the surface-to-wall gap δ unaffected and merely allows the computational domain to be restricted to the half-space $z \geq 0$, with a symmetry condition imposed on $\Gamma_{\text{sym}} = \{\mathbf{x} \in \Omega \setminus P : z = 0\}$,

$$u_z = 0 \quad \text{on } \Gamma_{\text{sym}}, \quad (18)$$

where u_z is the z -component of $\mathbf{u}$. The particle center therefore remains in the plane $z = 0$, and the motion reduces to the two in-plane components $x(t)$ and $y(t)$.

The second geometry is used for the particle-particle interactions of Section 4.2. In that case the single wall and fixed outer-boundary conditions are replaced by triply periodic boundary conditions on $\partial\Omega$, representing an infinite periodic array of particles.

3.4. Numerical method

The coupled velocity-pressure-conformation problem is solved with the finite-element method on unstructured, boundary-fitted tetrahedral meshes generated with Gmsh [38]. The velocity, pressure, and conformation tensor are approximated with P_2 - P_1 - P_1 elements; the P_2 - P_1 velocity-pressure pair satisfies the LBB condition. Because the mesh tracks the translating and rotating particle while the exterior boundaries remain fixed, an arbitrary Lagrangian-Eulerian (ALE) description is used [39]. The mesh velocity $\mathbf{u}_m$ is obtained from a Laplace problem for smooth mesh motion, and the material derivative is written as $D(\cdot)/Dt = \partial(\cdot)/\partial t|_{\mathbf{x}_m} + (\mathbf{u} - \mathbf{u}_m) \cdot \nabla(\cdot)$.

With the discretization in place, the remaining numerical ingredients are the standard stabilizations required for viscoelastic flow. The momentum balance is stabilized with the DEVSS-G formulation [40, 41]. The constitutive equation (14) is solved in the log-conformation representation [42, 43], which advances $\mathbf{s} = \log \mathbf{c}$ and reconstructs $\mathbf{c} = \exp(\mathbf{s})$, preserving positive definiteness. The convective transport in the constitutive equation is stabilized with the streamline-upwind/Petrov-Galerkin (SUPG) method [44].

4. Reduced-order models

This section derives the analytical reduced-order models used as forward maps in the Bayesian inference. Each model relates the applied force to the particle displacement while retaining only the hydrodynamic corrections needed for the setting under consideration. We proceed from Newtonian wall effects to Newtonian particle-particle interactions in a periodic domain, and finally to linear viscoelastic creep-recovery. In each case, the model starts from the Stokes mobility, adds the relevant correction, and is checked against the

finite-element full-order model of Section 3. Throughout, a is the particle radius and $h = a + \delta$ is the center-to-wall distance, with $\delta > 0$ the surface-to-wall gap.

4.1. Newtonian wall effects

In an unbounded creeping flow, the drag on a sphere translating with velocity $\mathbf{U}$ is given by Stokes' law, $\mathbf{F} = 6\pi\eta_s a \mathbf{U}$. A nearby rigid wall breaks this isotropy because the no-slip boundary reflects the disturbance flow back onto the particle. The resulting resistance depends on the dimensionless gap δ/a , is strongest as $\delta/a \rightarrow 0$, and vanishes as $\delta/a \rightarrow \infty$.

This wall correction is applied within the same in-plane geometry as the full-order problem. The force is prescribed as $\mathbf{F} = (F \cos \theta, F \sin \theta, 0)$ in the x - y plane, and the out-of-plane symmetry condition $u_z = 0$ at $z = 0$ keeps the particle center in that plane. The reduced-order model therefore tracks only the wall-parallel and wall-normal displacements $x(t)$ and $y(t)$.

The wall also introduces translation-rotation coupling: asymmetric traction on the particle surface can convert translation into rotation, especially for wall-parallel motion at small gaps. The full-order model resolves this coupling by solving for both $\mathbf{U}$ and the angular velocity $\boldsymbol{\omega}_p$ from the force and torque balance. The reduced-order model retains only the translational degrees of freedom. Its purpose is therefore to capture the dominant mobility change through direction-dependent resistance factors rather than to resolve the rotational dynamics explicitly.

With this reduction, the wall-corrected drag is written as

$$\mathbf{F} = 6\pi\eta_s a (f_{\parallel} U_{\parallel} \mathbf{e}_x + f_{\perp} U_{\perp} \mathbf{e}_y), \quad (19)$$

where $U_{\parallel}$ and $U_{\perp}$ are the wall-parallel and wall-normal velocity components, and $f_{\parallel}(\delta/a) \geq 1$ and $f_{\perp}(\delta/a) \geq 1$ are the corresponding resistance factors.

For perpendicular motion, the resistance factor is given by Brenner's bispherical-coordinate solution [45],

$$f_{\perp} = \frac{4}{3} \sinh \beta \sum_{n=1}^{\infty} \frac{n(n+1)}{(2n-1)(2n+3)} \left[\frac{2 \sinh((2n+1)\beta) + (2n+1) \sinh(2\beta)}{4 \sinh^2((n+\frac{1}{2})\beta) - (2n+1)^2 \sinh^2 \beta} - 1 \right], \quad (20)$$

where $\beta = \text{arccosh}(h/a)$. The series is truncated at $N_{\text{trunc}} = 100$ terms [46], which is sufficient for convergence down to the smallest gaps considered here. For parallel motion, we use the uniformly valid Zeng interpolant [47, 48],

$$f_{\parallel} \approx 1.028 - \frac{0.07 a^2}{a^2 + \delta^2} - \frac{8}{15} \ln \left(\frac{135 \delta}{135 a + 128 \delta} \right). \quad (21)$$

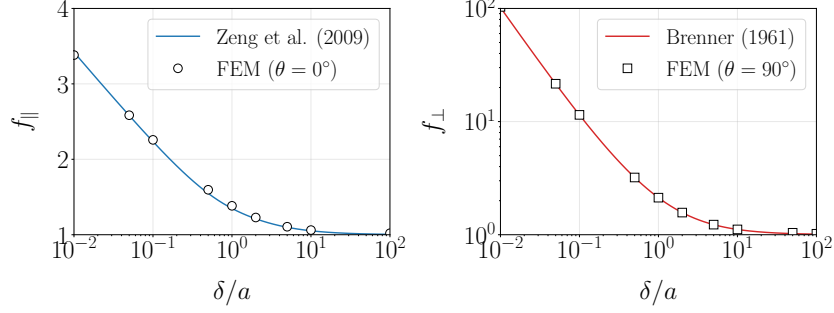


Figure 2: Wall-correction factors $f_{\parallel}$ and $f_{\perp}$ compared with finite-element reference data.

The two factors differ because the wall obstructs the two motions in different ways. Perpendicular motion displaces fluid through the thin gap between the particle and the wall, producing a lubrication resistance that grows as a/δ and therefore diverges algebraically as $\delta/a \rightarrow 0$. Parallel motion mainly shears the gap fluid, giving a weaker logarithmic divergence. Consequently, $f_{\perp} > f_{\parallel}$ at finite gaps, and the separation between the two corrections grows as the wall is approached. Both expressions agree with the finite-element reference data across the tested wall separations (Fig. 2).

As the particle moves, the wall factors are evaluated at the instantaneous separation $h(t) = a + \delta + y(t)$. A constant applied force therefore gives

$$\frac{dx}{dt} = \frac{F \cos \theta}{6\pi a \eta_s f_{\parallel}(h/a)}, \quad \frac{dy}{dt} = \frac{F \sin \theta}{6\pi a \eta_s f_{\perp}(h/a)}. \quad (22)$$

In the unbounded limit, $f_{\parallel} = f_{\perp} = 1$ and Eq. (22) reduces to Stokes' law. In bounded domains, oblique and perpendicular forcing excite the two corrected mobilities. Figure 3 shows that this two-factor model agrees closely with the finite-element displacements across the tested wall gaps and forcing angles.

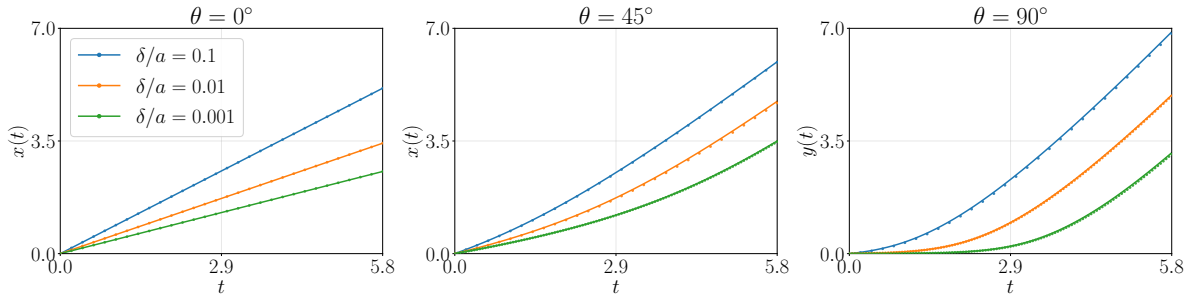


Figure 3: Reduced model (solid lines; Eq. (22)) versus finite-element particle displacements (dotted lines) for wall distances $\delta/a \in \{0.1, 0.01, 0.001\}$ and forcing angles $\theta \in \{0^\circ, 45^\circ, 90^\circ\}$. For each gap and forcing angle, the dotted FEM curves nearly coincide with the corresponding solid reduced-model curves, indicating excellent agreement across all tested cases.

4.2. Newtonian particle-particle effects

The wall correction accounts for a single rigid boundary. In a suspension, the probe also interacts hydrodynamically with neighboring particles. To isolate this second effect, we remove the wall and place the particle in a triply periodic domain, which represents an infinite periodic array of spheres. The forcing is applied only in the x direction, $\mathbf{F} = (F, 0, 0)$, so the reduced model tracks the corresponding translational velocity.

Hasimoto [49] showed that, relative to the unbounded Stokes drag $6\pi\eta_s a \mathbf{U}$, the drag in such a periodic lattice is rescaled by a dimensionless mobility ratio Q ,

$$\mathbf{F} = \frac{6\pi\eta_s a \mathbf{U}}{Q}, \quad (23)$$

which, for a simple cubic lattice of spacing L , depends only on the lattice geometry. With all lengths expressed in units of the particle radius a , so that L is the dimensionless spacing L/a ,

$$Q(L) = 1 - \frac{2.8373}{L} + \frac{4\pi}{3} \frac{1}{L^3} + \dots. \quad (24)$$

Because $Q < 1$, the drag scales as $1/Q$: periodic images reduce the particle mobility, so a larger force is required to obtain the same velocity as in an unbounded domain. As the spacing grows ($L \rightarrow \infty$), $Q \rightarrow 1$ and the isolated-particle Stokes drag is recovered.

We verify this correction against periodic-box finite-element simulations in Table 1, where all quantities are reported in non-dimensional form. The simulations use $F = 6\pi$, $\eta_s = 1$, and $a = 1$, so that the unbounded Stokes law gives a reference velocity $U_\infty = 1$. The simulated x -velocity U in the periodic box matches the Hasimoto prediction $Q(L)$ to better than 0.08% at every box size, and dividing it out recovers the unbounded reference $U_\infty = 1$ to within the same tolerance. This confirms that Eq. (23) captures the particle-particle contribution to the drag across the range of spacings considered.

L	U	$Q(L)$	U/Q	rel. error
10	0.7202479	0.7204588	0.999707	0.029%
20	0.8584315	0.8586586	0.999736	0.026%
40	0.9288854	0.9291329	0.999734	0.027%
60	0.9520053	0.9527311	0.999238	0.076%

Table 1: Verification of Hasimoto’s periodic Stokes drag. The raw velocity U matches the predicted factor $Q(L)$, and the corrected velocity U/Q recovers the unbounded value $U_\infty = 1$ to within 0.08%.

4.3. Viscoelasticity

The Newtonian models above are instantaneous: the particle velocity is set by the current force and mobility. A viscoelastic fluid adds memory, so the reduced model must also track the build-up and relaxation of polymeric stress. We consider a creep-recovery experiment in which a constant force of magnitude F at angle θ acts over $0 \leq t \leq t_0$ and is then removed, and we restrict the model to the linear viscoelastic regime $Wi \ll 1$. The construction proceeds in three steps. We first derive the unbounded creep-recovery model, then add wall effects through the Newtonian resistance factors of Section 4.1, and finally include periodic particle-particle effects through the Hasimoto factor of Section 4.2. Because the wall and periodic corrections modify only the mobility, both enter as simple modifications of the unbounded model.

4.3.1. Unbounded creep-recovery model

We begin in the unbounded domain, where neither a wall nor neighboring particles modify the mobility. The tensorial Oldroyd-B dynamics are reduced to a tractable set of ordinary differential equations by two approximations. First, the characteristic strain rate scales as $\dot{\gamma} \sim U/a$, so $Wi = \lambda \dot{\gamma} \sim \lambda U/a$; the order-unity prefactor implicit in this scaling is absorbed into the closure constant fixed below and contributes to the small residual calibration bias discussed in Section 6. Second, for $Wi \ll 1$, the convective and quadratic terms of the upper-convected derivative are subdominant. The polymeric stress can then be represented by a scalar magnitude $\tau_p(t) \equiv \|\boldsymbol{\tau}_p\|$, where $\|\cdot\|$ denotes the Frobenius norm. This scalar does not carry the sign or direction of the stress; those are restored below by assigning a separate relaxation equation to each Cartesian component of the polymeric force. Under these approximations, the Oldroyd-B law reduces to

$$\lambda \frac{d\tau_p}{dt} + \tau_p = \eta_p \frac{U}{a}, \quad (25)$$

in which the imposed particle velocity U drives the build-up of polymeric stress and λ sets the timescale over which it relaxes.

The scalar stress must next be converted into the polymeric contribution to the particle drag. We write this force as $F_p = C \tau_p$ and determine C by matching the steady-state Stokes drag associated with the polymer viscosity. Setting $d\tau_p/dt = 0$ in Eq. (25) gives the steady stress $\tau_p = \eta_p U/a$, while the corresponding Stokes polymer drag is $6\pi a \eta_p U$. Equating $C \tau_p = 6\pi a \eta_p U$ yields $C = 6\pi a^2$, so that

$$F_p = 6\pi a^2 \tau_p. \quad (26)$$

This closure reproduces the correct steady polymer drag. The short-time elastic response is governed by the modulus $G = \eta_p/\lambda$ through the same Eqs. (25)–(26), while the intermediate transient amplitude is captured approximately. Multiplying Eq. (25) by $6\pi a^2$ and using Eq. (26) gives the corresponding relaxation equation for the polymeric force,

$$\lambda \frac{dF_p}{dt} + F_p = 6\pi a \eta_p U. \quad (27)$$

The force relaxation is coupled back to the particle through the creeping-flow force balance. In the absence of inertia, the applied force is balanced at every instant by the sum of the solvent drag and the polymeric drag,

$$\underbrace{6\pi a \eta_s U}_{\text{solvent drag}} + \underbrace{F_p}_{\text{polymer drag}} = F(t). \quad (28)$$

The loading has wall-parallel and wall-normal components F_x and F_y . We therefore apply the scalar balance componentwise: the displacements are $x(t)$ and $y(t)$, the velocities are dx/dt and dy/dt , and the polymeric force components $F_{p,x}$ and $F_{p,y}$ obey separate copies of Eq. (27). Collecting the two displacement equations and the two relaxation equations gives the closed unbounded creep-recovery system

$$\begin{aligned} \frac{dx}{dt} &= \frac{1}{6\pi a \eta_s} [F_x(t) - F_{p,x}], & \frac{dy}{dt} &= \frac{1}{6\pi a \eta_s} [F_y(t) - F_{p,y}], \\ \frac{dF_{p,x}}{dt} &= -\frac{F_{p,x}}{\lambda} + \frac{6\pi a \eta_p}{\lambda} \frac{dx}{dt}, & \frac{dF_{p,y}}{dt} &= -\frac{F_{p,y}}{\lambda} + \frac{6\pi a \eta_p}{\lambda} \frac{dy}{dt}, \end{aligned} \quad (29)$$

with the initial conditions $F_{p,x}(0) = F_{p,y}(0) = 0$, reflecting a stress-free fluid at the onset of loading. The loading follows a standard creep-recovery protocol: a constant force of magnitude F at angle θ acts during $0 \leq t \leq t_0$ and is removed for $t > t_0$,

$$F_x(t) = \begin{cases} F \cos \theta, & 0 \leq t \leq t_0, \\ 0, & t > t_0, \end{cases} \quad F_y(t) = \begin{cases} F \sin \theta, & 0 \leq t \leq t_0, \\ 0, & t > t_0. \end{cases} \quad (30)$$

During the creep phase, the polymeric force builds up as the particle moves under load. During recovery, the applied force is removed and the stored polymeric force relaxes over the timescale λ , partially reversing the displacement. This unbounded system is the backbone of the viscoelastic surrogate. The wall and periodic-image corrections now enter by modifying its mobility.

4.3.2. Wall effects

Having established the unbounded viscoelastic model, we introduce the wall by reusing the Newtonian resistance factors $f_{\parallel}$ and $f_{\perp}$. For $Wi \ll 1$, the polymeric stress is only weakly perturbed by the boundary. We therefore assume that the same Newtonian factors, evaluated at the instantaneous gap $h(t) = a + \delta + y(t)$, multiply both the solvent and polymeric drag contributions. This approximation is consistent with Cho et al. [50], who found Faxén-type corrections accurate to a few percent in weakly elastic fluids. Under this assumption, the force balance (28) acquires a common wall factor in each direction,

$$\begin{aligned} 6\pi a \eta_s f_{\parallel} \frac{dx}{dt} + f_{\parallel} F_{p,x}(t) &= F_x(t), \\ 6\pi a \eta_s f_{\perp} \frac{dy}{dt} + f_{\perp} F_{p,y}(t) &= F_y(t). \end{aligned} \quad (31)$$

Dividing each line by its solvent-drag coefficient isolates the velocity. Because the same wall factor multiplies both drag terms in a given direction, it cancels from the polymeric-force term and remains only on the applied force. The polymeric-force relaxation is unchanged, since it comes from the constitutive law rather than the boundary condition. The wall-corrected system is therefore identical to Eq. (29) except that the applied force is divided by the appropriate wall factor,

$$\begin{aligned} \frac{dx}{dt} &= \frac{1}{6\pi a \eta_s} \left[\frac{F_x(t)}{f_{\parallel}} - F_{p,x} \right], & \frac{dy}{dt} &= \frac{1}{6\pi a \eta_s} \left[\frac{F_y(t)}{f_{\perp}} - F_{p,y} \right], \\ \frac{dF_{p,x}}{dt} &= -\frac{F_{p,x}}{\lambda} + \frac{6\pi a \eta_p}{\lambda} \frac{dx}{dt}, & \frac{dF_{p,y}}{dt} &= -\frac{F_{p,y}}{\lambda} + \frac{6\pi a \eta_p}{\lambda} \frac{dy}{dt}, \end{aligned} \quad (32)$$

with $F_{p,x}(0) = F_{p,y}(0) = 0$. Setting $f_{\parallel} = f_{\perp} = 1$ recovers the unbounded system (29), as expected. Figure 4 confirms close agreement with the FEM across $\delta/a \in \{0.1, 0.01, 0.005\}$ and $\theta \in \{0^\circ, 45^\circ, 90^\circ\}$.

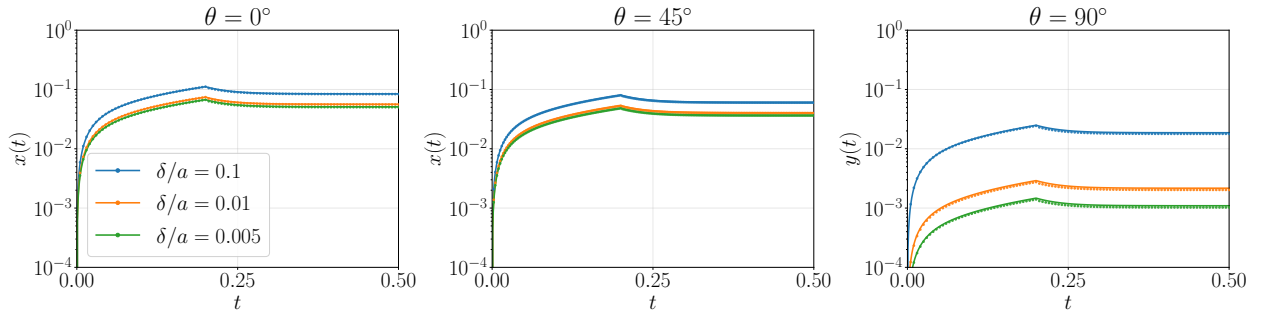


Figure 4: Reduced-order model (solid lines; Eq. (32)) versus finite-element particle displacements (dotted lines) for wall distances $\delta/a \in \{0.1, 0.01, 0.005\}$ and forcing angles $\theta \in \{0^\circ, 45^\circ, 90^\circ\}$. For each gap and forcing angle, the dotted FEM curves nearly coincide with the corresponding solid reduced-model curves, showing that the reduced model reproduces the finite-element creep-recovery response across all tested cases.

4.3.3. Particle-particle effects

The remaining correction accounts for the probe interacting with its periodic images. As in the Newtonian case, this interaction is represented by a scalar mobility factor. Following Section 4.2, the isotropic solvent mobility $1/(6\pi a \eta_s)$ in the viscoelastic system is replaced by $Q(L)/(6\pi a \eta_s)$, where $Q(L)$ is the Hasimoto factor of Eq. (24). Equivalently, this corresponds to setting $f_{\parallel} = f_{\perp} = 1/Q$ in Eq. (32). Dividing the periodic-box trajectory by $Q(L)$ then collapses the raw velocity and displacement onto the unbounded creep response, as shown in Fig. 5. Thus the same correction that recovers the Newtonian mobility also recovers the viscoelastic creep dynamics in the linear regime.

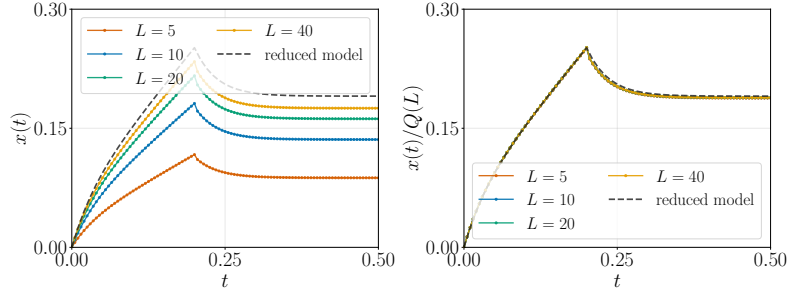


Figure 5: Periodic-box velocity $U(t)$ and displacement $x(t)$ (left) and the Hasimoto-corrected U/Q , x/Q (right) for box sizes $L \in \{10, 20, 30, 40, 60\}$; the corrected curves collapse onto the analytical (unbounded) prediction.

5. Inference procedure

The forward models of Section 4 define the deterministic map from the physical parameter vector $\boldsymbol{\mu}$ to the predicted particle trajectory $\mathbf{d}(\boldsymbol{\mu})$. For a Newtonian fluid, $\boldsymbol{\mu} = (\eta_s, \delta)^\top$; for a viscoelastic fluid, $\boldsymbol{\mu} = (\eta_s, \eta_p, \lambda, \delta)^\top$. The general Bayesian framework, including the observation model (3) and Bayes' theorem (4), was introduced in Section 2. We now specify the three ingredients needed to evaluate the posterior in practice: the prior distributions for the physical parameters and error-model hyperparameters (Section 5.1), the likelihood function (Section 5.2), and the MCMC sampler used to draw posterior samples (Section 5.3).

5.1. Prior distributions

The prior density encodes information available before the trajectory is observed. For the physical parameters, only positivity and broad physically reasonable ranges are assumed a priori. We therefore assign

log-uniform priors to the positive components of $\boldsymbol{\mu}$,

$$\begin{aligned}\log_{10} \eta_s &\sim \mathcal{U}(\log_{10} \eta_{s,\min}, \log_{10} \eta_{s,\max}), & \log_{10} \eta_p &\sim \mathcal{U}(\log_{10} \eta_{p,\min}, \log_{10} \eta_{p,\max}), \\ \log_{10} \lambda &\sim \mathcal{U}(\log_{10} \lambda_{\min}, \log_{10} \lambda_{\max}), & \log_{10} \delta &\sim \mathcal{U}(\log_{10} \delta_{\min}, \log_{10} \delta_{\max}).\end{aligned}\tag{33}$$

The relevant subset is used for each forward model. These priors are uniform over orders of magnitude rather than over the physical scale, so they do not concentrate probability near the upper end of a wide admissible interval. This choice follows the usual treatment of positive physical parameters whose uncertainty is multiplicative rather than additive: changing a viscosity from 0.1 to 1 carries the same order-of-magnitude meaning as changing it from 1 to 10, whereas a uniform prior on the physical scale would weight these intervals very differently.

The remaining unknowns describe the observation-error model. Following the Kennedy and O’Hagan formulation [34], we separate independent measurement noise from correlated model discrepancy. The experimental noise is modeled as

$$\boldsymbol{\varepsilon}_{\text{exp}} \sim \mathcal{N}(0, \sigma_{\text{exp}}^2 \mathbf{I}),\tag{34}$$

where $\sigma_{\text{exp}} > 0$ is the measurement-noise standard deviation and $\mathbf{I}$ is the identity matrix. The model-bias vector, evaluated at the measurement times, is represented by a zero-mean Gaussian process with a stationary exponential kernel [51]; in finite-dimensional form,

$$\boldsymbol{\varepsilon}_{\text{bias}} \sim \mathcal{N}(0, \sigma_{\text{bias}}^2 \mathbf{K}),\tag{35}$$

where $\sigma_{\text{bias}} > 0$ controls the amplitude of the correlated model discrepancy, and the kernel matrix $\mathbf{K}$ has entries

$$\mathbf{K}_{ij} = \exp\left(-\frac{|t_i - t_j|}{\ell_{\text{bias}}}\right).\tag{36}$$

Here ℓ_{bias} is the characteristic correlation timescale of the model discrepancy, measured in the same (non-dimensional) time units as t_i . The hyperparameters ℓ_{bias} , σ_{exp} , and σ_{bias} are inferred jointly with $\boldsymbol{\mu}$ and are assigned the priors

$$\ell_{\text{bias}} \sim \mathcal{U}(5\Delta t, t_{\text{final}}), \quad \sigma_{\text{exp}} \sim \mathcal{E}(\beta_{\text{exp}}), \quad \sigma_{\text{bias}} \sim \mathcal{E}(\beta_{\text{bias}}).\tag{37}$$

The lower bound on ℓ_{bias} prevents the correlated bias term from collapsing onto pointwise measurement

noise, while the upper bound prevents correlations longer than the observed trajectory. The exponential priors on σ_{exp} and σ_{bias} favor small error amplitudes while allowing larger values when supported by the data. The case-specific numerical values of these priors are reported with the experiments in Section 6.

5.2. Likelihood function

Since the experimental noise and model bias are both Gaussian, their sum is also Gaussian. The combined observation error $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_{\text{exp}} + \boldsymbol{\varepsilon}_{\text{bias}}$ therefore has zero mean and covariance matrix

$$\boldsymbol{\Sigma} = \sigma_{\text{exp}}^2 \mathbf{I} + \sigma_{\text{bias}}^2 \mathbf{K}, \quad (38)$$

where $\mathbf{I}$ is the identity matrix and $\mathbf{K}$ has entries given by Eq. (36). For two-dimensional trajectories $(x(t), y(t))$, the same notation $\boldsymbol{\Sigma}$ denotes the block-diagonal extension with one $N \times N$ block per displacement component. The blocks share the same kernel $\mathbf{K}$ and amplitudes σ_{exp} and σ_{bias} , reflecting the assumption that the wall-parallel and wall-normal errors are statistically independent but have the same noise and bias statistics.

The residual between the measured trajectory $\mathbf{y}_{\text{obs}}$ and the forward-model prediction $\mathbf{d}(\boldsymbol{\mu})$ is

$$\mathbf{r}(\boldsymbol{\mu}) = \mathbf{y}_{\text{obs}} - \mathbf{d}(\boldsymbol{\mu}), \quad \mathbf{r}(\boldsymbol{\mu}), \mathbf{y}_{\text{obs}}, \mathbf{d}(\boldsymbol{\mu}) \in \mathbb{R}^m, \quad (39)$$

where $m = N$ for a single displacement component and $m = 2N$ for a two-component trajectory. For two-component data, $\mathbf{y}_{\text{obs}}$ collects the measured pairs $(x(t_i), y(t_i))$, and $\mathbf{d}(\boldsymbol{\mu})$ collects the corresponding forward-model predictions. The likelihood is then the probability density of the combined error evaluated at this residual,

$$p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\mu}, \ell_{\text{bias}}, \sigma_{\text{exp}}, \sigma_{\text{bias}}) = \frac{1}{(2\pi)^{m/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2} \mathbf{r}^\top \boldsymbol{\Sigma}^{-1} \mathbf{r}\right), \quad (40)$$

where m is the dimension of the stacked trajectory vector. The corresponding log-likelihood is

$$\log p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\mu}, \ell_{\text{bias}}, \sigma_{\text{exp}}, \sigma_{\text{bias}}) = -\frac{1}{2} \mathbf{r}^\top \boldsymbol{\Sigma}^{-1} \mathbf{r} - \frac{1}{2} \log |\boldsymbol{\Sigma}| - \frac{m}{2} \log(2\pi). \quad (41)$$

Both the quadratic form and the log-determinant are evaluated via a Cholesky factorization of $\boldsymbol{\Sigma}$, avoiding explicit matrix inversion and improving numerical stability.

5.3. Posterior sampling

With the priors (33)–(37) and the likelihood (40) defined, Bayes' theorem (4) fully specifies the posterior. Let $\boldsymbol{\vartheta} = (\boldsymbol{\mu}, \ell_{\text{bias}}, \sigma_{\text{exp}}, \sigma_{\text{bias}})^\top$ collect the physical parameters, the bias correlation timescale, and the two

error amplitudes. Since the evidence $p(\mathbf{y}_{\text{obs}})$ is independent of $\boldsymbol{\vartheta}$, it contributes only an additive normalization constant to the log posterior. The target density is therefore evaluated as

$$\log p(\boldsymbol{\vartheta} \mid \mathbf{y}_{\text{obs}}) = \log p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\vartheta}) + \log p(\boldsymbol{\vartheta}) + C, \quad (42)$$

where $C = -\log p(\mathbf{y}_{\text{obs}})$ is omitted in the numerical implementation. Because the forward models are nonlinear in the inferred parameters and the covariance contains inferred hyperparameters, this posterior is not available in closed form and must be characterized numerically.

We employ the affine-invariant ensemble MCMC sampler of Goodman and Weare [52], as implemented in `emcee` [53]. The sampler evolves an ensemble of N_{w} walkers in parallel and uses the current ensemble geometry to construct proposals. Its affine invariance reduces sensitivity to linear rescaling and correlations in parameter space, which is useful for the anisotropic posteriors produced by the wall and viscoelastic cases.

All components of $\boldsymbol{\vartheta}$ are positive, so the walkers are evolved in base-10 logarithmic coordinates,

$$\boldsymbol{\phi} = \log_{10} \boldsymbol{\vartheta}, \quad \boldsymbol{\vartheta} = 10^{\boldsymbol{\phi}}. \quad (43)$$

The physical-parameter priors in Eq. (33) are already defined on this $\log_{10}$ scale. Priors specified on the physical scale, including the uniform prior on ℓ_{bias} and the exponential priors on the error amplitudes, are transformed with the corresponding change-of-variables Jacobian. Besides matching the log-uniform priors above, this parameterization avoids non-physical negative proposals and makes the stretch move act on relative changes in scale, which is more efficient when admissible values span several orders of magnitude. Let $\pi(\boldsymbol{\phi})$ denote the posterior density in the transformed variables.

At each iteration, the ensemble state is $\{\boldsymbol{\phi}^{(j)}\}_{j=1}^{N_{\text{w}}}$. Proposals are generated using the stretch move. For walker j , a complementary walker $\boldsymbol{\phi}^{(\ell)}$, with $\ell \neq j$, is selected uniformly from the remaining walkers, and the candidate state is

$$\boldsymbol{\phi}^* = \boldsymbol{\phi}^{(\ell)} + Z(\boldsymbol{\phi}^{(j)} - \boldsymbol{\phi}^{(\ell)}), \quad (44)$$

where Z is drawn from $g(z) \propto z^{-1/2}$ on $[1/a_{\text{s}}, a_{\text{s}}]$, with $a_{\text{s}} = 2$. The candidate is accepted with probability

$$\gamma = \min \left(1, Z^{N_{\text{dim}}-1} \frac{\pi(\boldsymbol{\phi}^*)}{\pi(\boldsymbol{\phi}^{(j)})} \right), \quad (45)$$

where N_{dim} is the dimension of the transformed sampling space. The factor $Z^{N_{\text{dim}}-1}$ is the stretch-move

volume factor required to preserve detailed balance with respect to the target posterior. It is separate from the change-of-variables Jacobian above: the Jacobian enters the target density $\pi(\phi)$, whereas $Z^{N_{\text{dim}}-1}$ belongs to the proposal mechanism.

In all experiments, we use $N_w = 2N_{\text{dim}}$ walkers and run 10,000 iterations per walker. The first 30% of each chain is discarded as burn-in, corresponding to the initial period in which walkers move from their prior initialization toward the typical set of the posterior. Convergence is then assessed by inspecting trace plots to verify that the walkers mix across the same stationary region rather than remaining separated by their initial positions, and by computing the Gelman-Rubin statistic [54, 55] from the ensemble chains. Chains are accepted as converged when $\hat{R}$ is close to one, in practice $\hat{R} \lesssim 1.1$, and when posterior means, credible intervals, and correlations are stable with respect to the discarded burn-in.

6. Numerical experiments

We apply the inference framework to a sequence of numerical experiments of increasing complexity, using finite-element solutions as surrogate measurement data throughout. The experiments first establish what can be identified from an isolated probe, then add wall-induced mobility corrections, and finally include particle-particle effects in periodic suspensions. Across these cases, the constitutive response advances from Newtonian to linear viscoelastic behavior and then to regimes where the linear surrogate begins to fail. Table 2 summarizes the forcing, geometry, and true parameters for each case.

Case	Fluid	Forcing F, θ	Geometry (δ/a)	True parameters
Isolated (§6.1)	Newtonian	$12\pi, 0^\circ$	unbounded	$\eta_s = 1$
Isolated	Oldroyd-B	$8\pi, 0^\circ$	unbounded	$\eta_s = 0.5, \eta_p = 0.9, \lambda = 0.1$
Nonlinear	Oldroyd-B	$800\pi, 0^\circ$	unbounded	$\eta_s = 0.5, \eta_p = 0.9, \lambda = 0.1$
Multi-force (linear)	Oldroyd-B	$\{16, 32, 64\}\pi, 0^\circ$	unbounded	as above
Multi-force (high-Wi)	Oldroyd-B	$\{200, 400, 800\}\pi, 0^\circ$	unbounded	as above
Wall	Newtonian	$12\pi, \{0^\circ, 45^\circ\}$	$\{0.001, 0.01, 0.1\}$	$\eta_s = 1$
Wall	Oldroyd-B	$8\pi, 45^\circ$	$\{0.005, 0.01, 0.1\}$	$\eta_s = 0.5, \eta_p = 0.9, \lambda = 0.1$
Particle-particle	Oldroyd-B	$6\pi, 0^\circ$	periodic (L)	$G = \eta_p/\lambda = 9$

Table 2: Overview of the numerical experiments, listing for each case the fluid type, the applied force magnitude and angle, the geometry (wall gap δ/a or periodic box), and the true material parameters. The probe radius is $a = 1$ throughout. Noise-sensitivity studies add synthetic Gaussian noise to the isolated Newtonian and Oldroyd-B cases.

6.1. Isolated particle

6.1.1. Newtonian fluid

We begin with an isolated particle of radius $a = 1$ driven by a constant force $F = 12\pi$ at an angle $\theta = 0^\circ$ through an unbounded Newtonian fluid with true viscosity $\eta_s = 1$. With no wall present, the only unknown

physical parameter is η_s , so this one-parameter problem provides a clean baseline before wall effects are introduced.

Parameter	Prior
$\log_{10} \eta_s$	$\mathcal{U}(\log_{10}(0.01), \log_{10}(100))$
$\log_{10} \delta$	$\mathcal{U}(\log_{10}(0.001), \log_{10}(100))$
ℓ_{bias}	$\mathcal{U}(1, 5.8)$
σ_{exp}	$\mathcal{E}(\beta_{\text{exp}})$
σ_{bias}	$\mathcal{E}(100)$

Table 3: Prior distributions used for the Newtonian inference cases.

Using the priors in Table 3, we infer η_s from the displacement trajectory via MCMC. Synthetic zero-mean Gaussian noise is added to the FEM trajectory, with standard deviation σ_{noise} prescribed as a fixed fraction of the maximum displacement $u_{\text{max}} = \max_t |u(t)|$; for the posterior in Fig. 6, this fraction is 2%. The posterior is sharply concentrated around the true value, with posterior mean $\hat{\eta}_s = 1.0$, confirming that viscosity is strongly identifiable in the unbounded Newtonian regime.

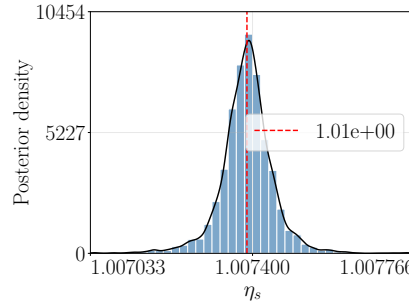


Figure 6: Marginal posterior distribution of η_s for the isolated Newtonian particle with synthetic noise standard deviation 2% of u_{max} .

We next examine how the prescribed noise level influences the inference, considering scales of 0.5%, 2%, 5%, and 10% of u_{max} . The corresponding noisy trajectories are shown in Fig. 7, and the posterior summaries in Table 4. The inferred viscosity is robust: even at 10% noise, $\hat{\eta}_s$ stays within about 1% of the true value. Increasing the noise mainly broadens the posterior and raises the inferred experimental-noise standard deviation $\hat{\sigma}_{\text{exp}}$, both roughly in proportion to the added noise. This robustness follows from the simplicity of the Newtonian response: the displacement grows linearly in time, so the inference reduces to estimating a single slope, and zero-mean scatter about that line averages out over the trajectory.

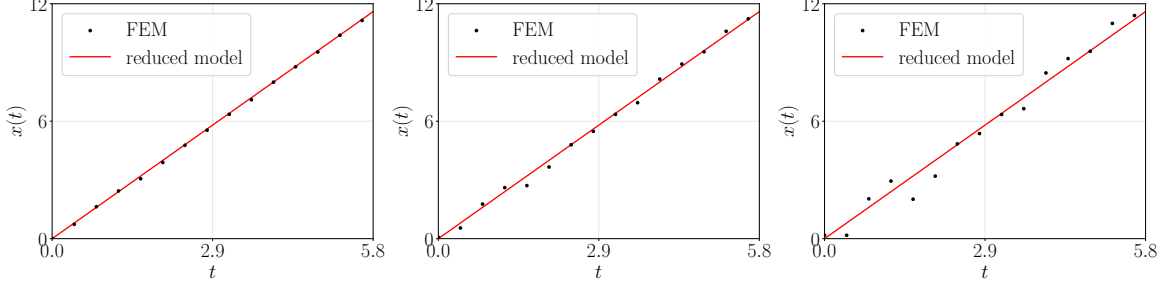


Figure 7: Newtonian displacement response with synthetic Gaussian noise added to the FEM data. The noise is zero-mean with standard deviation σ_{noise} chosen as a percentage of the maximum displacement $u_{\text{max}} = \max_t |u(t)|$: 0.5% in the left panel, 2% in the middle panel, and 5% in the right panel. The red curve denotes the reduced model prediction, and the black dots denote the FEM simulations.

Noise	$\hat{\eta}_s$	$\text{std}(\hat{\eta}_s)$	$\hat{\sigma}_{\text{exp}}$	$\text{std}(\hat{\sigma}_{\text{exp}})$
0.5%	1.0	2.36×10^{-3}	5.64×10^{-2}	1.11×10^{-2}
2%	1.0	8.64×10^{-3}	2.30×10^{-1}	4.40×10^{-2}
5%	1.00	2.06×10^{-2}	5.71×10^{-1}	1.15×10^{-1}
10%	1.01	4.18×10^{-2}	1.15	2.14×10^{-1}

Table 4: Effect of the prescribed noise percentage on inference for the Newtonian unbounded model. The estimate $\hat{\eta}_s$ stays within about one percent of the true value $\eta_s = 1$ at all noise levels, while the inferred experimental-noise standard deviation $\hat{\sigma}_{\text{exp}}$ (posterior mean) grows in proportion to the added noise.

6.1.2. Viscoelastic fluid

The Newtonian baseline tests whether a single slope identifies a single material parameter. We next add memory by considering an Oldroyd-B fluid, where the same trajectory must identify both viscous and elastic time-dependent contributions.

Parameter	Prior
$\log_{10} \eta_s$	$\mathcal{U}(\log_{10}(0.01), \log_{10}(100))$
$\log_{10} \eta_p$	$\mathcal{U}(\log_{10}(0.01), \log_{10}(100))$
$\log_{10} \lambda$	$\mathcal{U}(\log_{10}(0.01), \log_{10}(100))$
$\log_{10} \delta$	$\mathcal{U}(\log_{10}(0.001), \log_{10}(100))$
ℓ_{bias}	$\mathcal{U}(0.05, 0.5)$
σ_{exp}	$\mathcal{E}(\beta_{\text{exp}})$
σ_{bias}	$\mathcal{E}(100)$

Table 5: Prior distributions used for the viscoelastic inference cases.

A particle of radius $a = 1$ is driven by a constant force $F = 8\pi$ at $\theta = 0^\circ$ over $0 \leq t \leq t_0 = 0.2$; the force is then removed and the simulation continues to $t = 0.5$ to capture the full creep-recovery response. The fluid is Oldroyd-B with true parameters $\eta_s = 0.5$, $\eta_p = 0.9$, and $\lambda = 0.1$. The broad log-uniform priors listed in Table 5 encode weak prior knowledge of the rheological parameters.

For the posterior in Fig. 8, synthetic zero-mean Gaussian noise is added with standard deviation set to

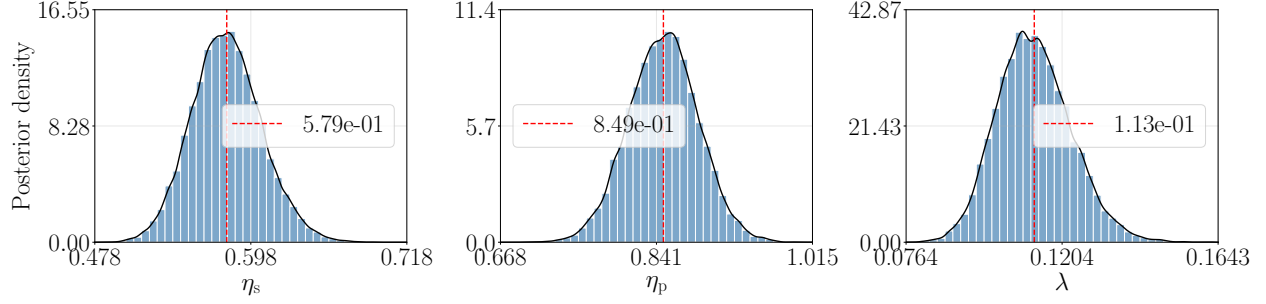


Figure 8: Marginal posterior distributions of $(\eta_s, \eta_p, \lambda)$ for the unbounded-domain creep-recovery case with true parameters $\eta_s = 0.5$, $\eta_p = 0.9$, $\lambda = 0.1$. The synthetic noise standard deviation is 2% of the maximum displacement at $t = t_0 = 0.2$.

2% of the maximum displacement, attained at the end of forcing at $t = t_0 = 0.2$. All three parameters are reasonably well constrained, with the true value of each lying within the posterior uncertainty. A single creep-recovery trajectory therefore contains enough information to jointly identify $(\eta_s, \eta_p, \lambda)$ within the uncertainty set by the prescribed noise. This is possible because each parameter governs a distinct feature of the response: η_s controls the initial creep rate, λ the timescale of stress build-up and relaxation, and η_p the magnitude of the elastic contribution during recovery.

As in the Newtonian case, we test the sensitivity of the inference to measurement noise by adding zero-mean Gaussian noise to the FEM trajectory at 0.5%, 2%, 5%, and 10% of the maximum displacement $u_{\max}$ (Fig. 9), with the resulting posteriors reported in Table 6. The estimates remain accurate up to about 2% noise, where $(\eta_s, \eta_p, \lambda)$ are still recovered close to their true values with a small spread. At 5%, however, the inference degrades sharply: η_p and λ drift well away from their true values and their standard deviations grow by roughly two orders of magnitude; by 10%, the estimates are no longer meaningful. This contrast reflects the greater complexity of the creep-recovery response relative to the Newtonian case.

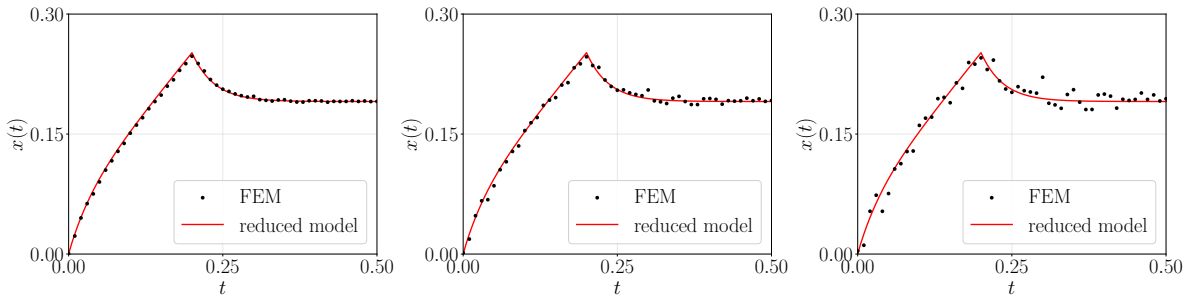


Figure 9: Viscoelastic displacement response with synthetic Gaussian noise added to the FEM data. The noise is zero-mean with standard deviation σ_{noise} chosen as a percentage of the maximum displacement $u_{\max} = \max_t |u(t)|$: 0.5% in the left panel, 2% in the middle panel, and 5% in the right panel. The red curve denotes the reduced model prediction, and the black dots denote the FEM simulations.

Noise	$\hat{\eta}_s$	$\text{std}(\hat{\eta}_s)$	$\hat{\eta}_p$	$\text{std}(\hat{\eta}_p)$	$\hat{\lambda}$	$\text{std}(\hat{\lambda})$	$\hat{\sigma}_{\text{exp}}$	$\text{std}(\hat{\sigma}_{\text{exp}})$
0.5%	0.549	1.38×10^{-2}	0.857	1.42×10^{-2}	1.03×10^{-1}	3.95×10^{-3}	1.08×10^{-3}	1.92×10^{-4}
2%	0.557	2.67×10^{-2}	0.843	2.35×10^{-2}	1.03×10^{-1}	6.39×10^{-3}	3.92×10^{-3}	4.08×10^{-4}
5% [†]	0.635	1.55×10^{-1}	1.37	1.79	0.747	1.93	9.69×10^{-3}	1.17×10^{-3}
10% [†]	1.04	1.24×10^{-1}	5.72	2.64	6.14	2.50	1.93×10^{-2}	2.37×10^{-3}

Table 6: Effect of the prescribed noise percentage on inference for the unbounded creep-recovery model; $\hat{\sigma}_{\text{exp}}$ is the posterior mean of the inferred experimental-noise standard deviation. At 5% and 10% noise, the posterior standard deviations of η_p and λ equal or exceed their means: the polymer parameters are no longer identified, and the corresponding point estimates should not be interpreted as material properties.

6.1.3. Nonlinear viscoelastic

All preceding viscoelastic experiments were run at moderate Weissenberg numbers, within the range of validity of the linear surrogate. To locate where this approximation breaks down, Fig. 10a first scales the forcing relative to the baseline creep case, with $F = 8\pi c$ (so that $c = 1$ corresponds to $F = 8\pi$) and the displacement scaled by c . Up to approximately $c = 32$ ($F = 256\pi$), the scaled trajectories nearly collapse, indicating an approximately linear response; for larger forcing, the collapse is lost, marking the onset of nonlinear viscoelastic effects. We therefore test a strongly nonlinear case with $F = 800\pi$ and $\theta = 0^\circ$, for which $\text{Wi} \gg 1$ and the linear creep model is not expected to be quantitatively accurate. Figure 10b

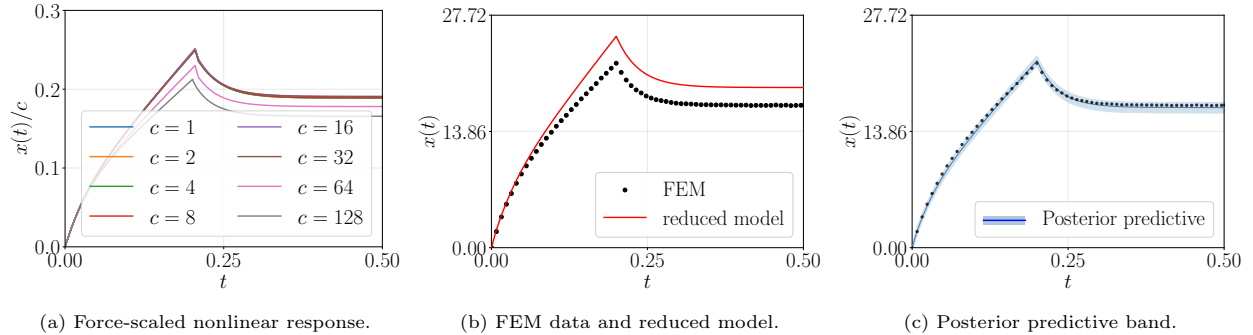


Figure 10: High-Weissenberg-number creep test at $F = 800\pi$. The reduced linear surrogate underpredicts the nonlinear FEM peak displacement, but a single-trajectory posterior predictive fit can partially mask this structural error by shifting the inferred parameters.

compares the FEM trajectory with the reduced-order prediction in this high- Wi regime. The surrogate captures the early-time response reasonably well, where the motion is governed mainly by viscous resistance. At later times, however, nonlinear elastic effects become important: the FEM trajectory reaches a larger peak displacement, and the reduced model underestimates both the peak displacement and the relaxation amplitude.

For a single trajectory, this structural error can be partly hidden by the posterior predictive fit. The band in Fig. 10c closely follows the FEM data because the inferred parameters shift to compensate for the

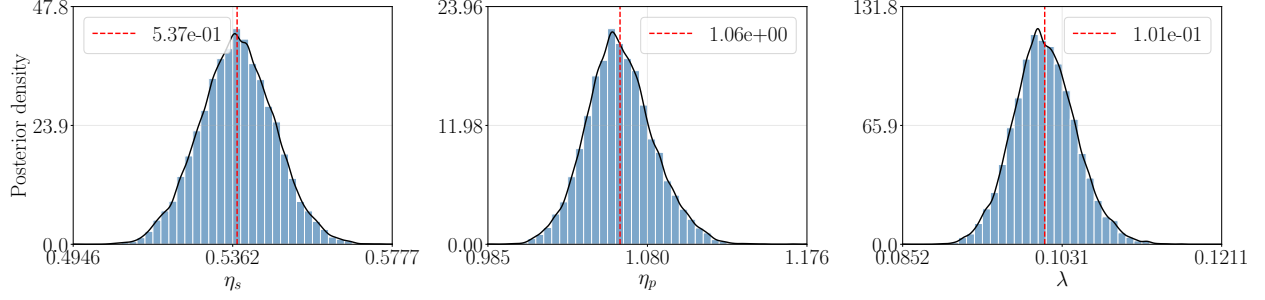


Figure 11: Marginal posterior distributions of $(\eta_s, \eta_p, \lambda)$ for the nonlinear creep test at $F = 800\pi$ ($Wi \gg 1$). Vertical dashed lines mark the true parameter values. The estimated η_p drifts upward to absorb the unrepresented nonlinear elastic contribution.

missing nonlinear physics. This is evident in Fig. 11: η_s and λ remain close to their true values, while η_p shifts upward ($\eta_p^{\text{true}} = 0.9$ versus $\hat{\eta}_p \approx 1.06$), absorbing the nonlinear elastic contribution that the surrogate cannot represent.

The single-trajectory fit is therefore not a reliable diagnostic of model adequacy. To expose the model error, we fit several force levels jointly, so that one biased parameter set cannot compensate for the missing nonlinearities at all forces simultaneously. We compare a lower-force set, $F \in \{16\pi, 32\pi, 64\pi\}$, which stays in the linear-response regime, with a high-force set, $F \in \{200\pi, 400\pi, 800\pi\}$, which enters the high- Wi regime.

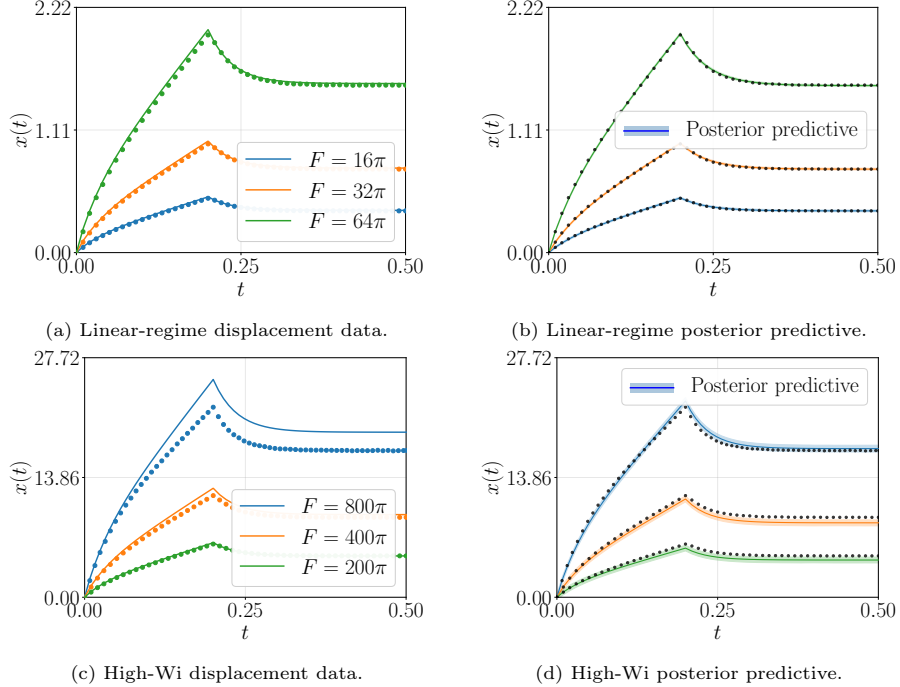


Figure 12: Joint posterior predictive checks for mixed creep-recovery data. The left panels show the displacement curves, and the right panels show the posterior predictive bands. The lower-force set $F \in \{16\pi, 32\pi, 64\pi\}$ is described well by the surrogate, whereas the high-force set $F \in \{200\pi, 400\pi, 800\pi\}$ shows systematic deviations near peak displacement and during recovery, revealing the limit of the linear viscoelastic surrogate.

Figure 12 shows the resulting posterior predictive checks. For the lower-force set, the surrogate describes all three trajectories well. For the high-force set, systematic discrepancies grow with F , especially near the peak displacement and during recovery. The joint fit therefore reveals that the discrepancy is not merely a poor point estimate but evidence of unrepresented nonlinear physics in the forward model.

6.2. Wall effects

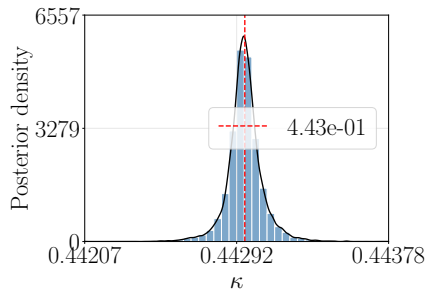
6.2.1. Newtonian fluid

We now introduce a wall and first consider wall-parallel motion. In the wall-corrected Stokes relation, the resistance depends on the product $\eta_s f_{\parallel}(\delta/a)$, so increasing the viscosity or decreasing the wall distance reduces the particle mobility in the same way. From a single wall-parallel trajectory, these two effects cannot be separated: many pairs (η_s, δ) produce the same displacement slope.

For this parallel-motion case, the identifiable quantity is the effective mobility parameter

$$\kappa = \frac{1}{f_{\parallel}(\delta/a) \eta_s}, \quad (46)$$

which captures the combined effect of viscosity and wall-induced parallel resistance. For trajectories generated with $F = 12\pi$ at $\theta = 0^\circ$ and $\delta/a = 0.1$, the reference effective mobility is $\kappa = 0.443$. As shown in Fig. 13, the posterior concentrates around this value, confirming that κ is well identified.



(a) $\delta/a = 0.1$.

Figure 13: Posterior distribution of the effective mobility κ in the bounded domain at $\delta/a = 0.1$.

The single-direction experiment thus fixes only the parallel mobility κ , providing a single observable, the slope $\dot{x}$, and cannot separate η_s from δ . Oblique forcing breaks this degeneracy by producing both wall-parallel and wall-perpendicular motion: any angle $\theta \in (0^\circ, 90^\circ)$ yields two independent slopes,

$$\dot{x} = \frac{F \cos \theta}{6\pi a \eta_s f_{\parallel}(\delta/a)}, \quad \dot{y} = \frac{F \sin \theta}{6\pi a \eta_s f_{\perp}(\delta/a)}, \quad (47)$$

whose ratio

$$\frac{\dot{y}}{\dot{x}} = \tan \theta \frac{f_{\parallel}(\delta/a)}{f_{\perp}(\delta/a)} \quad (48)$$

depends only on δ/a , since the viscosity cancels. The two resistance factors diverge at different rates as $\delta/a \rightarrow 0$ ($f_{\parallel}$ logarithmically, $f_{\perp}$ algebraically), so their ratio $f_{\parallel}/f_{\perp}$ varies strongly over the near-wall range considered. The slope ratio therefore provides direct information about δ , after which either component of Eq. (47) fixes η_s . The choice $\theta = 45^\circ$ used here balances the sensitivities of $\dot{x}$ and $\dot{y}$ and is adopted for definiteness; the argument applies equally to any $\theta \in (0^\circ, 90^\circ)$.

We now examine whether η_s and δ can be jointly identified from a bounded-domain trajectory generated with oblique forcing, $F = 12\pi$ at $\theta = 45^\circ$, $\delta/a = 0.1$, and synthetic noise standard deviation 2% of $u_{\max}$. Figure 14 shows the marginal posteriors and the correlation between η_s and δ . The posterior uncertainty contains the true value of each parameter, the posterior means lie close to them, and the off-diagonal panel shows a strong positive correlation between the two.

The preceding results use the wall-corrected forward model. To isolate the effect of model error, Fig. 15 instead fits the unbounded Newtonian model to bounded-domain FEM data at $\delta/a = 0.01$. For wall-parallel forcing ($\theta = 0^\circ$), the posterior predictive still follows the data closely, but only by inferring biased parameters that compensate for the omitted wall drag. At $\theta = 45^\circ$ this compensation becomes less effective and the band widens; for wall-normal forcing ($\theta = 90^\circ$), where the wall correction is strongest, it widens further and the mismatch becomes most apparent. The posterior predictive thus reveals the growing structural error as the motion becomes more wall-normal.

Table 7 reports posterior means and standard deviations for η_s and δ at three true gap distances, with $\eta_s = 1$ and baseline noise. In every case the posterior means are close to the true values and the posterior uncertainty contains the truth, including at the smallest gap $\delta = 0.001$, where wall effects are strongest.

True δ	Parameter	Mean	Std
0.001	η_s	1.004	1.7×10^{-3}
	δ	0.001	1.9×10^{-5}
0.01	η_s	0.9987	6.8×10^{-4}
	δ	0.0097	7.2×10^{-5}
0.1	η_s	0.9984	6.8×10^{-4}
	δ	0.095	7.9×10^{-4}

Table 7: Posterior mean and standard deviation for η_s and δ at three true wall distances, with fixed $\eta_s = 1$ at baseline noise.

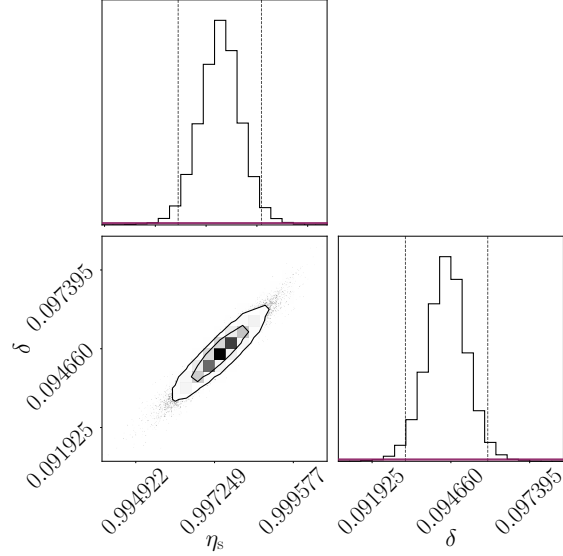


Figure 14: Marginal posterior distributions and correlation between η_s and δ for the bounded-domain Newtonian case at $\delta/a = 0.1$. Diagonal panels show the 1D marginal posteriors; dashed vertical lines mark the 95% credible interval, and the red curves show the prior densities in the plotted coordinates. The off-diagonal panel shows the correlation between the two parameters, with contours enclosing the 68% and 95% credible regions.

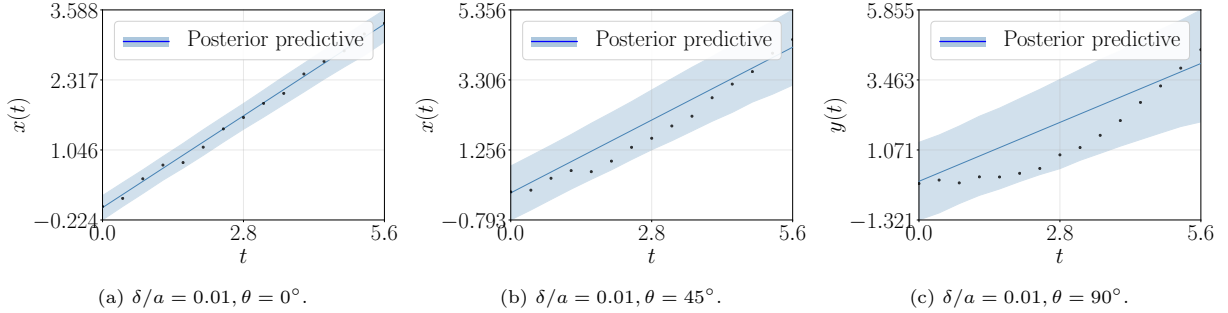


Figure 15: Posterior predictive bands obtained when the unbounded Newtonian model is fit to bounded-domain FEM data at $\delta/a = 0.01$. The mismatch increases from wall-parallel forcing to wall-normal forcing, consistent with the increasing importance of the omitted wall correction.

6.2.2. Viscoelastic fluid

Having analyzed isolated creep-recovery trajectories, we now add the wall separation as a fourth unknown. This bounded viscoelastic problem combines the two identifiability mechanisms seen above: the transient shape separates $(\eta_s, \eta_p, \lambda)$, while the oblique trajectory separates the wall-parallel and wall-normal mobilities needed to infer δ . We therefore use $\theta = 45^\circ$, so that both mobility components are excited in the same experiment.

For the representative case $\delta/a = 0.1$, Fig. 16 shows the marginal and joint posteriors of $(\eta_s, \eta_p, \lambda, \delta)$. The diagonal panels show unimodal marginals concentrated well inside the broad prior ranges, indicating that the trajectory is informative about all four unknowns. The off-diagonal panels reveal a clear correlation

between the wall separation δ and the two viscosities η_s and η_p , since both a smaller gap and larger viscosities reduce mobility. By contrast, λ remains nearly uncorrelated with δ , indicating that the inferred relaxation time is largely unaffected by wall effects.

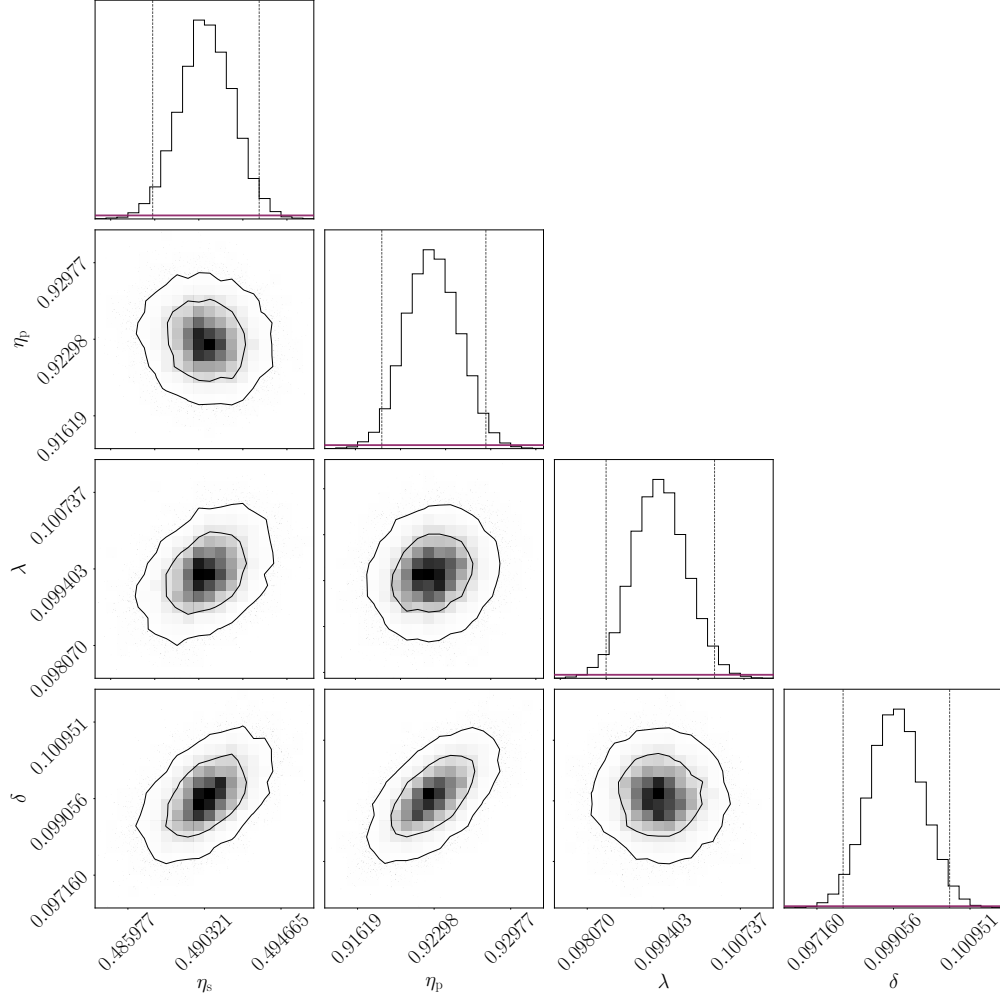


Figure 16: Marginal and joint posterior distributions of $(\eta_s, \eta_p, \lambda, \delta)$ for the bounded-domain creep-recovery case at $\delta/a = 0.1$. Diagonal panels show the 1D marginal posteriors; dashed vertical lines mark the 95% credible interval. Off-diagonal panels show the 2D joint posteriors, with contours enclosing the 68% and 95% credible regions.

Table 8 reports the posterior means and standard deviations for $(\eta_s, \eta_p, \lambda, \delta)$ at three true wall distances, $\delta \in \{0.005, 0.01, 0.1\}$. The posterior means stay close to the true values across the tested gaps, but a small systematic bias is visible: the solvent viscosity is underestimated by about 6 to 8%, and the polymer viscosity is overestimated by up to about 7%. The relaxation time and wall separation are recovered more tightly. This small offset is consistent with the steady-drag closure $C = 6\pi a^2$ introduced in Section 4.3.1: matching the polymeric drag at steady state leaves the transient creep amplitude only approximately represented, so

True δ	Parameter	True value	Mean	Std
0.005	η_s	0.50	0.46	1.5×10^{-2}
	η_p	0.90	0.92	3.1×10^{-2}
	λ	0.10	0.095	1.4×10^{-3}
	δ	0.005	0.0049	1.1×10^{-3}
0.01	η_s	0.50	0.46	7.1×10^{-3}
	η_p	0.90	0.92	1.2×10^{-2}
	λ	0.10	0.098	1.1×10^{-3}
	δ	0.010	0.0097	7.3×10^{-4}
0.1	η_s	0.50	0.47	5.2×10^{-3}
	η_p	0.90	0.96	8.8×10^{-3}
	λ	0.10	0.094	1.3×10^{-3}
	δ	0.100	0.100	2.4×10^{-3}

Table 8: Posterior mean and standard deviation at three true gap distances, with true parameters $\eta_s = 0.5$, $\eta_p = 0.9$, and $\lambda = 0.1$.

the inference compensates by trading a slightly lower η_s against a slightly higher η_p . The discrepancy is small and correlated in time, so the Gaussian-process bias term absorbs the residual without compromising the identifiability of the four parameters.

6.3. Particle-particle interactions

The wall corrections of the previous sections isolate a single probe from one boundary. In a suspension, however, the probe also feels its neighbors. We model this second interaction through the periodic Hasimoto correction of Section 4.2 and test it in two settings: a homogeneous material, where the modulus does not vary in space and the correction can be verified directly, and a heterogeneous material, where the local modulus varies in space and single-particle probing is compared against a bulk measurement.

6.3.1. Homogeneous material

In the homogeneous material, the true modulus is prescribed as $G_{\text{true}} = 9$. When the Hasimoto correction is omitted, the finite size of the periodic box biases the inferred modulus. Table 9 shows that G is strongly overestimated for small domains: the raw inference returns $G_{\text{inferred}} = 19.71$ at $L = 5$, more than twice the true value. The error falls steadily as the box grows, reaching about 7% at $L = 60$. This residual offset decays only algebraically with box size, because the periodic images interact through a slowly decaying Stokes field; this is precisely the contribution that the Hasimoto factor removes. Applying the correction is therefore what makes single-particle inference reliable, and we carry the same correction into the spatially varying material considered next.

L	G_{inferred}	$\text{std}(G_{\text{inferred}})$	Error (%)
5	19.71	0.770	119
10	12.71	0.480	41.2
20	10.67	0.407	18.6
30	10.09	0.380	12.1
40	9.83	0.369	9.2
60	9.63	0.390	7.0

Table 9: Effect of domain size on the inferred modulus in a homogeneous material with $G_{\text{true}} = 9$ when the Hasimoto correction is *not* applied. The error is the relative deviation of G_{inferred} from G_{true} .

6.3.2. Heterogeneous material

We now turn to a material in which the modulus varies in space, $G = G(\mathbf{x})$, while the relaxation time λ is held fixed and uniform. The construction of the heterogeneous microstructure in the full-order model is described in Section Appendix B. Here the field is represented as a uniform background G_0 plus a superposition of $N_b = 20$ Gaussian blobs,

$$G(\mathbf{x}) = G_0 + \sum_{i=1}^{N_b} A_i \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_i\|_2^2}{2w_i^2}\right), \quad (49)$$

where each blob width w_i equals its radius R_i . The radius and amplitude of each blob are drawn independently from $R_i \in \{0.5, 1.0, 1.5, 2.0\}$ and $A_i \in \{2, 5, 10, 20\}$, and the N_b centers $\mathbf{c}_i$ are placed by Latin-hypercube sampling for even spatial coverage. With $G_0 = 9.0 = \eta_p/\lambda$, the background reproduces the homogeneous reference case, and $L_x = L_y = L_z = 10$. Because the blob widths satisfy $w_i \ll L$, the periodic images of each Gaussian contribute negligibly at the box center, so G_0 is recovered as the background modulus away from the peaks. Figure 17(a) shows the resulting isosurfaces of G , and panel (b) shows planar slices at $z = -2.5, 0, +2.5$: smooth, well-separated peaks ($G \sim 30\text{--}34$) sit on a lower background ($G \sim 10\text{--}13$).

The central question in a heterogeneous material is what a single moving probe actually measures: the pointwise modulus at its location, a local spatial average over the region it disturbs, or the bulk modulus of the material. Because the probe stress samples the field over a region set by the particle size, we expect it to report a local, weighted effective modulus rather than the pointwise value $G(\mathbf{x}_p)$. We address this by comparing three quantities: the distribution of local estimates G_{loc} obtained from many probe configurations, their mean $\bar{G}_{\text{loc}}$, and an independent bulk modulus G_{bulk} from macroscopic shear.

The probe interacts with its own periodic images exactly as in the homogeneous case, so the same finite-size correction of Section 4.2 removes the periodic-box artifact before the particle response is converted into an apparent modulus. Each local estimate is obtained from the full posterior, with the modulus $G = \eta_p/\lambda$

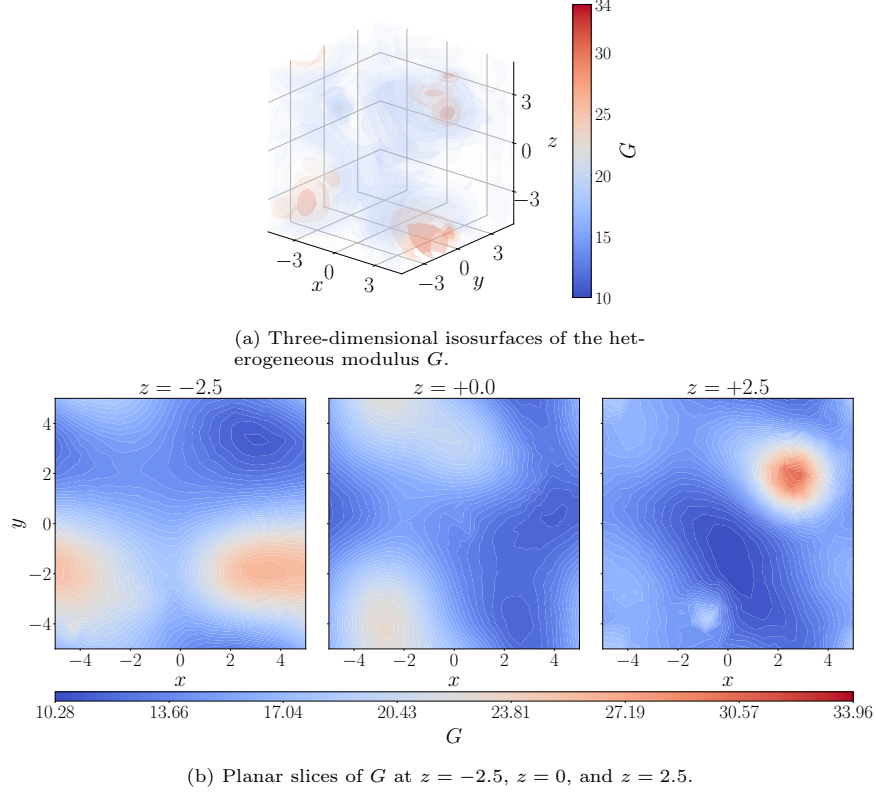


Figure 17: Visualization of the heterogeneous modulus field G . The isosurfaces highlight the three-dimensional heterogeneity, while the slices show the spatial variation through the domain.

evaluated from the posterior samples of η_p and λ rather than from their means alone, so that the reported G_{loc} carries the inference uncertainty. To sample the field without regenerating the particle mesh, the particle is held fixed at the box center and the entire periodic field is translated rigidly by a random offset $\mathbf{s} = (s_x, s_y, s_z)$, with each component drawn uniformly on $[0, L)$ and wrapped periodically. This is equivalent to evaluating $G(\mathbf{x} + \mathbf{s})$ at the fixed particle location. A new offset is drawn for each of $N_{\text{run}} = 40$ independent runs, giving 40 independent, Hasimoto-corrected local estimates $G_{\text{loc}}^{(k)}$ of the shear modulus. Their histogram (Fig. 18) has mean

$$\bar{G}_{\text{loc}} = \frac{1}{40} \sum_{k=1}^{40} G_{\text{loc}}^{(k)} = 17.44. \quad (50)$$

As a reference, the same field is probed in bulk, with no particle present. A homogeneous simple shear is imposed through Lees–Edwards periodic boundary conditions: a velocity jump $\Delta u = u_x(\text{top}) - u_x(\text{bottom})$ across the periodic y -faces sets a macroscopic shear rate $\dot{\gamma} = \Delta u / L_y$, and the volume-averaged shear stress $\langle \sigma_{xy} \rangle$ is tracked in time. Starting from a stress-free state, the first strain increment $\Delta \gamma = \dot{\gamma} \Delta t$ probes the purely elastic, pre-relaxation response of the material, so the bulk modulus is the secant slope of stress over

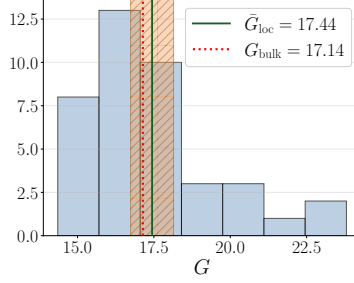


Figure 18: Histogram of the 40 Hasimoto-corrected local estimates $G_{\text{loc}}^{(k)}$ (mean $\bar{G}_{\text{loc}} = 17.44$), shown with the 95% confidence interval and the bulk reference $G_{\text{bulk}} = 17.14$.

strain at this first step,

$$G_{\text{bulk}} = \frac{\Delta\langle\sigma_{xy}\rangle}{\Delta\gamma} = 17.14. \quad (51)$$

The 95% confidence interval of $\bar{G}_{\text{loc}}$ brackets G_{bulk} , so the two measurements are statistically consistent in the mean. What each reveals, however, is different. G_{bulk} is a single number, blind by construction to the spatial structure of Fig. 17: it would be unchanged if the material were exactly uniform at 17.14. The particle-resolved estimates instead recover the full distribution, whose spread is direct evidence of the underlying heterogeneity. This answers the question posed above: the single-particle probe measures neither the pointwise modulus nor the bulk value, but a local effective modulus. Its scatter reflects the spatial variation of the field. Bulk rheology alone therefore gives an incomplete picture of the material, whereas Hasimoto-corrected single-particle probing resolves both the local modulus and its distribution.

7. Conclusions

The central theme of this work is that active microrheology becomes quantitatively reliable only when the systematic biases introduced by the measurement environment are corrected and the residual uncertainty is propagated into the inferred parameters. Left uncorrected, wall drag, finite periodic-domain effects, particle-particle interactions, heterogeneous sampling, and viscoelastic model inadequacy at high Weissenberg number each distort the inferred rheology. We have investigated whether the transient creep-recovery trajectory of a force-controlled sphere translating near a rigid wall can be used to infer, in a single experiment, the solvent viscosity, polymer viscosity, and relaxation time of a viscoelastic fluid, together with the particle-to-wall separation, once these corrections are in place. Within the moderate-Wi regime where the linearized viscoelastic surrogate remains valid, the answer is affirmative. The Bayesian formulation further provides principled diagnostics, through posterior predictive checks and an explicit Gaussian-process bias model, to

identify when the experiment has entered a regime where the surrogate no longer adequately describes the observations.

The main findings are as follows. (i) For Newtonian fluids, both the viscosity η_s and the particle-to-wall separation δ are identifiable from two-directional force-controlled trajectories, while the unbounded Newtonian baseline remains robust even when synthetic noise reaches 10% of the maximum displacement. In bounded geometries, the naturally identifiable quantity from a single-direction trajectory is the combined mobility parameter $\kappa = [f_{\parallel}(\delta/a)\eta_s]^{-1}$; separating η_s and δ requires two-directional forcing data or an independent geometric measurement. (ii) For linear viscoelastic fluids, creep-recovery trajectories contain sufficient information to jointly identify the three constitutive parameters and, under oblique forcing, the wall separation. The joint posterior reflects the Oldroyd-B coupling structure, including the correlations between wall separation and viscosity, and between η_p and λ . (iii) At large Weissenberg number ($Wi \gg 1$), a single trajectory can be fit by biased effective parameters, but joint posterior predictive checks across multiple force levels expose the structural mismatch. (iv) For periodic particle-particle interactions, the Hasimoto correction removes the dominant finite-domain mobility artifact, enabling single-particle estimates to recover the distribution of a heterogeneous modulus field rather than only its bulk average.

Practical guidance on the applicable regime can be summarized as a validity map in the control parameters Wi , the dimensionless gap δ/a , and the dimensionless box size L/a . For $Wi \ll 1$, moderate gaps, and sufficiently large L/a , the linear surrogate can be applied directly, and the posterior uncertainty provides a meaningful measure of parameter identifiability. As Wi approaches or exceeds unity, or as the gap becomes very small, the bias term and posterior predictive checks become essential diagnostics rather than optional uncertainty estimates. For $Wi \gg 1$, posterior predictive failure should be interpreted as a cue to adopt a nonlinear forward model or an accurate emulator, and parameters inferred with the linear-regime surrogate should be treated as effective qualitative descriptors only. At small L/a the Hasimoto correction is required to remove the finite-domain bias before any modulus is reported.

Future directions include (i) multi-mode relaxation spectra, replacing the single-mode Maxwell surrogate with a discrete sum of relaxation modes; (ii) nonlinear constitutive inference (*e.g.* Giesekus model), enabled by surrogate models or emulators that reduce the cost of repeated forward evaluations within MCMC; and (iii) application to experimental particle-tracking data, where careful treatment of model bias and instrumental systematics is essential.

Appendix A. Mesh and domain-size convergence studies

Two sources of discretization error are assessed for both Newtonian and viscoelastic settings: (i) the spatial element size (mesh convergence) and (ii) the size of the truncated computational box (domain-size convergence). Both studies are reported for the three-dimensional (3D) bounded geometry and the axisymmetric unbounded geometry.

The mesh hierarchies and domain-size levels are summarized in Tables A.10 and A.11. Here, h_{particle} denotes the characteristic element size near the particle surface and h_{far} denotes the characteristic size in the far field.

For the 3D bounded geometry, the convergence of the Newtonian and viscoelastic solutions is shown in Figs. A.19 and A.21, respectively. For the axisymmetric unbounded geometry, the corresponding results are given in Figs. A.20 and A.22.

The particle displacement is found to converge at mesh M3 on domain D2 in the 3D geometry ($\delta/a = 0.1$) and at mesh M2 on domain D2 in the axisymmetric geometry for both Newtonian and viscoelastic cases. These discretizations are adopted as the reference configuration for all subsequent simulations.

Mesh	h_{particle}	h_{far}	nodes	elements
M1	0.8	48	1 242	540
M2	0.4	24	3 555	1 827
M3	0.2	12	13 918	8 380
M4	0.1	6	85 766	57 644

(a) 3D mesh hierarchy on the reference domain D2.

Domain	L	W	H
D1	50	50	50
D2	100	100	100
D3	200	200	200
D4	400	400	400

(b) Domain-size levels for the 3D geometry.

Table A.10: Spatial discretization and truncation levels for the 3D mesh and domain-size convergence studies.

Mesh	h_{particle}	h_{far}	nodes	elements
M1	1.6	14	586	261
M2	0.8	8	1 894	887
M3	0.4	4	6 785	3 274
M4	0.2	2	25 651	12 590

(a) Axisymmetric mesh hierarchy on the reference domain D2.

Domain	R_c	ℓ_{c1}	ℓ_{c2}
D1	50	50	50
D2	100	100	100
D3	200	200	200
D4	400	400	400

(b) Domain-size levels for the axisymmetric geometry.

Table A.11: Spatial discretization and truncation levels for the axisymmetric mesh and domain-size convergence studies.

Appendix A.1. Newtonian

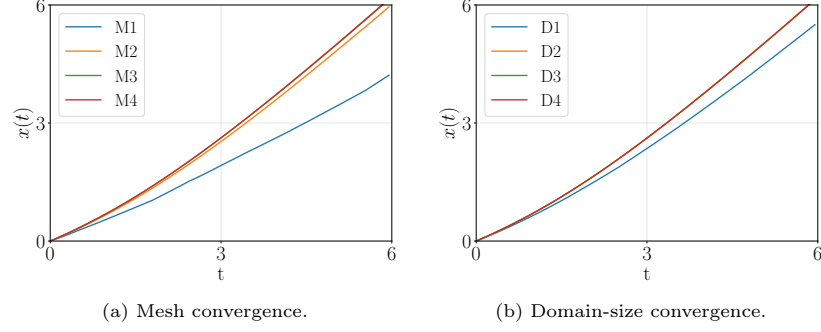


Figure A.19: 3D bounded geometry ($\delta/a = 0.1$): (a) effect of mesh refinement on domain D2 and (b) effect of domain size for the converged mesh (Newtonian fluid).

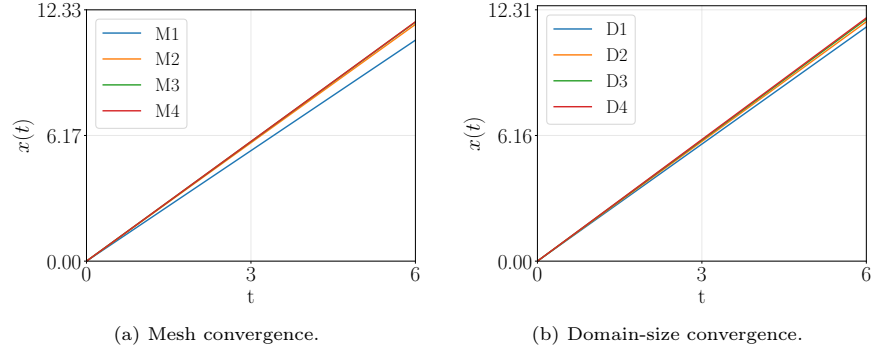


Figure A.20: Axisymmetric unbounded geometry: (a) effect of mesh refinement on domain D2 and (b) effect of domain size for the converged mesh (Newtonian fluid).

Appendix A.2. Viscoelastic

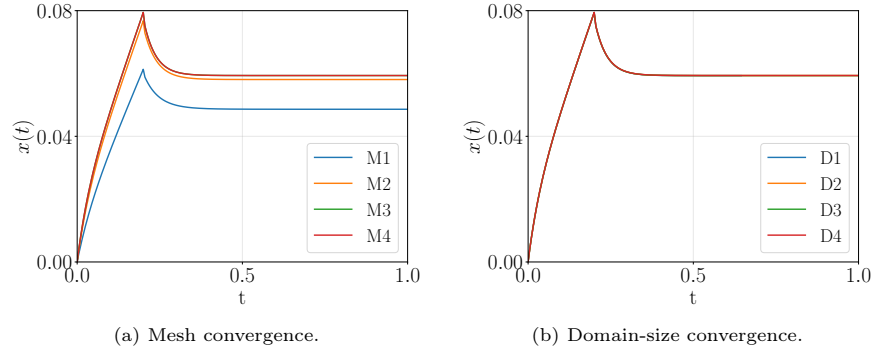


Figure A.21: 3D bounded geometry ($\delta/a = 0.1$): (a) effect of mesh refinement on domain D2 and (b) effect of domain size for the converged mesh (viscoelastic fluid).

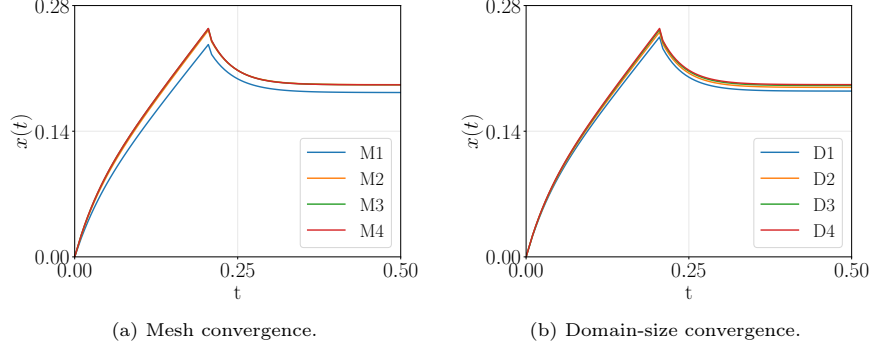


Figure A.22: Axisymmetric unbounded geometry: (a) effect of mesh refinement on domain D2 and (b) effect of domain size for the converged mesh (viscoelastic fluid).

Appendix B. Modeling the heterogeneous microstructure

The models so far assume a spatially homogeneous fluid. To probe a material whose elasticity varies in space (Section 6.3.2), we now equip the full-order model with a transported microstructure whose parameters we likewise aim to infer. We introduce a scalar concentration field $\phi(\mathbf{x}, t)$ that modulates the local elasticity of the fluid. It is initialized as a collection of spherical blobs seeded at randomly placed centers $\mathbf{x}_k$, each contributing a non-truncated Gaussian profile,

$$\phi_k(\mathbf{x}) = \phi_{\text{blob}} \exp\left(-\frac{r_k^2}{2w^2}\right), \quad r_k = \|\mathbf{x} - \mathbf{x}_k\|, \quad (\text{B.1})$$

where ϕ_{blob} is the peak concentration and w is its width. The centers are placed at random and may overlap; their number is fixed by a prescribed nominal volume fraction.

The field is then transported by the flow, satisfying the following advection equation

$$\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi = 0 \quad \text{in } \Omega \setminus P, \quad (\text{B.2})$$

discretized with the same second-order semi-implicit scheme as the flow problem. The concentration enters the constitutive model through the local modulus, while the relaxation time is kept fixed:

$$G(\phi) = \frac{\eta_p}{\lambda} \phi, \quad \lambda(\phi) \equiv \lambda, \quad (\text{B.3})$$

so that the microstructure changes only through the modulus. Since the polymeric stress is $\boldsymbol{\tau}_p = G(\phi)(\mathbf{c} - \mathbf{I})$, the varying $G(\phi)$ is evaluated point-wise at every Gauss point from the convected field, giving a fluid whose local elasticity tracks the transported microstructure and a particle that experiences a heterogeneous,

evolving viscoelastic environment.

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